# 1 Probability

## 1.1 Experiment

> An experiment is any activity or process whose outcome is subject to uncertainty. …
>
> Thus experiments that may be of interest include tossing a coin once or several times, selecting a card or cards from a deck, weighing a loaf of bread, ascertaining the commuting time from home to work on a particular morning, obtaining blood types from a group of individuals, or measuring the compressive strengths of different steel beams. (Devore p 51)

> An experiment is any process, real or hypothetical, in which the possible outcomes can be identified ahead of time. (DeGroot p 5)

Note that different experiments can be assigned to an activity:

- Experiment: Flip a coin 2x and record the result of each flip. Can ask what fraction of experiments had a head on the first flip.
- Experiment: Flip a coin 2x and record the number of heads and tails. Can ask what fraction of experiments had one head.

## 1.2 Outcome

The result of part of an experiment or the result of the full experiment. (Not generally defined; the first definition implies it is the result of the full experiment, which is inconsistent with the implicit definition in the first definition.)

## 1.3 Sample Space, $\mathcal{S}$

> … the set of all possible outcomes of an experiment. (Devore p 51)

> The collection of all possible outcomes of an experiment is called the *sample space* of the experiment. (DeGroot p 6)

The outcomes in a sample space are noted using the shorthand

$\mathcal{S} = \{\text{outcome 1}, \text{outcome 2}, \ldots\}$

For example, if the experiment is tossing a coin twice and writing down the result of the first toss in box 1 and the result of the second toss in box 2,

$\mathcal{S} = \{HH, HT, TH, TT\}$

If the experiment is tossing a coin twice and counting the total number of heads and tails, the three simple events are $1H1T$, $2T$, $2H$, and the sample space is

$\mathcal{S} = \{1H1T, 2T, 2H\}$

For both experiments, define $A$ as the event of obtaining exactly one tail. It is compound in the first experiment, consisting of $HT$ and $TH$, and simple in the second experiment, consisting of $1H1T$.

*Question*: Experiment: Each day, shoot free throws until you miss. What are the outcomes that make up $\mathcal{S}$?

**Demonstration**

Simulate the experiment of shooting a free throw 10 times. Assume you make 80% of your free throws.

*Question*: How would you use the simulation to estimate the chances that you get 10 in a row? Start with the sample code that follows.

Partial answer to a simpler problem.

```
# Simulate the experiment of shooting a free throw 2 times.
# Assume you make 50% of your free throws.

# How would you use the simulation to estimate the chances that you get 10 in a row?
# Start with native Python library. Will consider better options later.
import random
#a = [1, 1, 1, 1, 1, 1, 1, 1, 0, 0]
a = [0, 1]
# Randomly select an element from the list a
result = random.choice(a)
n_e = 100000 # Number of experiments
n_t = 2 # Number of tosses per experiment

m = 0 # Number of experiments when we make all shots
for experiment in range(n_e):
  n = 0
  for toss in range(n_t):
    result = random.choice(a)
    if result == 1:
      n = n + 1
  if n == n_t:
    m = m + 1
    print(f"Experiment {experiment+1:d}: {n_t} in a row")

print(m/n_e)
```

*Question*: How many experimental outcomes are in $\mathcal{S}$? If you execute this number of experiments with your code, will all experimental outcomes in $\mathcal{S}$ have been generated?

*Question*: Describe how you would use a program to print all outcomes.

## 1.4 Event

> An event is any collection (subset) of outcomes contained in the sample space $\mathcal{S}$. An event is simple if it consists of exactly one outcome and compound if it consists of more than one outcome. (Devore p 52)

> An event is a well-defined set of possible outcomes of the experiment. (DeGroot p 5)

*Question*: Can we also say a sample space is the set of all possible simple events?

## 1.5 Set

> The sample space of an experiment can be thought of as a *set*, or a collection, of different possible outcomes; and each outcome can be thought of as a *point*, or an *element*, in the sample space. Similarly, events can be thought of as subsets of the sample space. (DeGroot p 6)

## 1.6 *Problem*

Describe an activity that requires the use of the terms “Experiment”, “Outcome”, “Sample Space”, and “Event”. Bonus for not using coin tosses. Double bonus if funny.

## 1.7 *Demonstration and Prelude to Error Bars and Hypothesis Tests*

The following program draws 10 random numbers to form list $x$ and 10 random numbers to form list $y$ and plots the results. Ordinary least squares regression is used to find the line of best fit. We are interested in the outcome that the best fit slope is greater than 0.1.

*Question*: What is the experiment? What are outcomes in $\mathcal{S}$? What are the events?

Code. We will do many such experiments on homework.

```
import numpy as np
import matplotlib.pyplot as plt

# Generate random data on interval [0, 1]
x = np.random.rand(10)
print(x)
y = np.random.rand(10)
print(y)

# Perform ordinary least squares regression
# We'll cover this later in the semester.
# Solve y = m*x + c for the best-fit slope m and intercept c
A = np.vstack([x, np.ones(len(x))]).T
print(A)

fit = np.linalg.lstsq(A, y, rcond=None)
print(fit)

m, c = fit[0]

# Discuss issues with this plot and my expectations for homework submissions.
plt.scatter(x, y, label='Data points')
plt.plot(x, m*x + c, 'r', label='Best fit line')
plt.xlabel('x')
plt.ylabel('y')
plt.legend()
plt.show()
```

## 1.8 Definition of Probability

The term probability refers to the study of randomness and uncertainty. In any situation in which one of a number of possible outcomes may occur, the discipline of probability provides methods for quantifying the chances, or likelihoods, associated with the various outcomes. (Devore p 50)

Given an experiment and a sample space $\mathcal{S}$, the objective of probability is to assign to each event $A$ a number $P(A)$, called the probability of the event $A$, which will give a precise measure of the chance that $A$ will occur. (Devore p 55)

> A statistical probability is thus the limiting value of the relative frequency with which some event occurs. (Bulmer p 4)

> Probability will be the way that we quantify how likely something is to occur (in the sense of one of the interpretations in Sec. 1.2. [Frequency Interpretation, Classical Interpretation, and Subjective Interpretation]). (DeGroot p 5)

## 1.9 Replication

A repetition of an experiment.

## 1.10 Relative Frequency and Interpretation of Probability

> The interpretation [of probability] most frequently used and most easily understood is based on the notion of relative frequencies. (Devore p 57)

Repeat the experiment $n$ times (each repetition is called a "replication"). If event $A$ occurs $n(A)$ times in $n$ replications, then relative frequency is $n(A)/n$.

> The *objective interpretation of probability* identifies this limiting relative frequency to $P(A)$ (Devore p 57)

Said another way,

$$P(A) = \lim_{n \to \infty} \frac{n(A)}{n}$$

If an experiment is not repeatable, prior information must be used to determine $P(A)$ and not everyone may conclude the same $P(A)$; in this case, $P(A)$ has a subjective interpretation. See DeGroot 2012 and Ross 2022 for a discussion of interpretations of probability.

Python example of Figure 2.2 Devore:

```
import random

a = [0, 1]

# Experiment: Randomly select an element from the list a
result = random.choice(a)

# Repeat the experiment n times
n = 2
results = []
for exp in range(1, n+1):
  result = random.choice(a)
  results.append(result)

print(f"n = {n} experiments:")
print(f"  Results: {results}")
print(f"  rf(0) = {results.count(0) / n}")
print(f"  rf(1) = {results.count(1) / n}")
```

## 1.11 Set Operators

### 1.11.1 Complement

"Not $A$" is represented by four symbols: $A'$ $\overline{A}$ $\sim A$ $\neg A$

> The complement of an event $A$, denoted by $A'$, is the set of all outcomes in $\mathcal{S}$ that are not contained in $A$. (Devore p 53)

### 1.11.2 Difference

$A - B$ means the events in which $A$ occurs but not $B$.

### 1.11.3 Subset (or containment)

$A \subset B$ means the set $A$ is a subset of $B$.

### 1.11.4 Union

"Or" (union; inclusive or) is typically represented by $\cup$

> The union of two events $A$ and $B$, denoted by $A \cup B$ "$A$ or $B$" is the event consisting of all outcomes that are *either in $A$ or in $B$ or in both events* (so that the union includes outcomes for which both $A$ and $B$ occur as well as outcomes for which exactly one occurs) – that is, all outcomes in at least one of the events. (Devore p 53)

### 1.11.5 XOR

XOR – "Exclusive or": $A \oplus B$ means the event that is in $A$ or $B$, but not both.

### 1.11.6 Intersection

"And" (intersection) is represented by $\cap$. An ampersand (`&`) or comma may also denote "and" in context, but they are not universal set-intersection symbols.

> The intersection of two events $A$ and $B$, denoted by $A \cap B$ and read "$A$ and $B$," is the event consisting of all outcomes that are in *both* $A$ *and* $B$. (Devore p 53)

### 1.11.7 In Python

See Python notes

## 1.12 Null Event

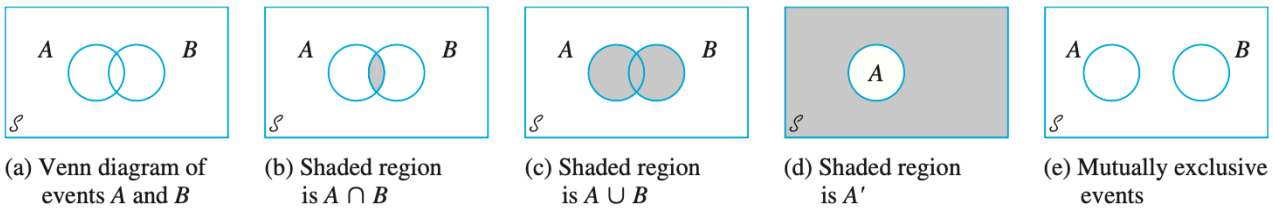
> Let $\varnothing$ denote the null event (the event consisting of no outcomes whatsoever). When $A \cap B = \varnothing$, $A$ and $B$ are said to be mutually exclusive or disjoint events. (Devore p 54)

## 1.13 Mutually Exclusive

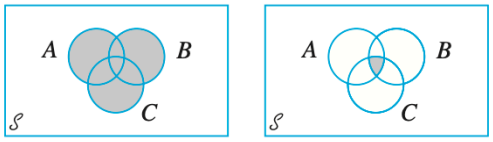
Defined implicitly in Null Event definition. Also referred to as "pairwise disjoint".

## 1.14 Venn Diagrams

Venn diagrams are useful for visually describing set operators. It is debatable if they are the best option for describing the relationships between sets for much else.




(a) Venn diagram of events $A$ and $B$ (b) Shaded region is $A \cap B$ (c) Shaded region is $A \cup B$ (d) Shaded region is $A'$ (e) Mutually exclusive events




(f) Shaded region is $A \cup B \cup C$ (g) Shaded region is $A \cap B \cap C$

Drawing the Venn diagram for an experiment with 2 flips where $A$ is one or more heads and $B$ is one or more tails.

Use set notation to describe the region of $A$ that is not shaded in Figure (b).

Use set notation to describe the region outside of $A$ and $B$ in Figure (a).

## 1.15 Axioms of Probability

Often referred to as Kolmogorov's Axioms

1. For any event $A$, $P(A) \geq 0$.
2. $P(\mathcal{S}) = 1$
3. If $A_1, A_2, \ldots$ is an infinite collection of disjoint events, then

$$P(A_1 \cup A_2 \ldots) = P\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} P(A_i)$$

Corollary

$$P(A_1 \cup A_2 \ldots . \cup A_k) = \sum_{i=1}^{k} P(A_i)$$

Note that many textbooks give the third axiom in terms of a sum over two events instead of a sum over an infinite set of events.

(Devore p 56)

Axioms do not completely determine an assignment of probabilities to events. The axioms serve only to rule out assignments inconsistent with our intuitive notions of probability. (Devore p 57)

Corollary to Axiom 3 (Devore p 59 calls this a proposition):

For any event $A$, $P(A) \leq 1$

**Example**

In independent trials where the result is either true (with probability $1 - p$) or false (with probability $p$), with $0 < p \leq 1$, we run trials until we get a false. The sample space of all experiments is $\mathcal{S} = \{A_1, A_2, A_3, \ldots\}$, where

$A_1 = F, A_2 = T, F, A_3 = T, T, F, \ldots$

By Axioms 2 and 3, we expect

$P(A_1) + P(A_2) + P(A_3) + ....$

to equal 1 because the $A_i$s are disjoint. We will learn later why $P(A_1) = p$, $P(A_2) = (1-p)p$, $P(A_3) = (1-p)^2 p$, …

Using this, we have

$P(A_1) + P(A_2) + P(A_3) + .... = p + (1-p)p + (1-p)^2 p + ...$

Recall the formula for an infinite geometric series:

$$a + ar + ar^2 + ... = \frac{a}{1-r} \qquad |r| < 1$$

With $a = p$ and $r = 1 - p$, we conclude

$P(A_1) + P(A_2) + P(A_3) + .... = 1$

See also Devore Example 2.12.

## 1.16 Law of Complements

(Devore does not use this term, however)

> For any event $A$, $P(A) + P(A') = 1$, from which $P(A) = 1 - P(A')$. (Devore p 59)

## 1.17 Law of Addition

(Devore does not use this term, however)

For any two events $A$ and $B$ that are mutually exclusive,

$P(A \cup B) = P(A) + P(B)$

## 1.18 General Law of Addition

(Devore does not use this term, however)

> For any two events $A$ and $B$,
> $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

> For any three events $A$, $B$, and $C$,
> $P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(A \cap C) - P(B \cap C) + P(A \cap B \cap C)$

**Visual proof for two events**

Imagine overlapping targets $A$ and $B$, and darts are thrown towards the targets.

Visually, the number of ways $A$ or $B$ occurred: $n(A \cup B) = n(A) + n(B) - n(A \text{ and } B)$

Divide by the total number of dots, $n$, use the relative frequency interpretation of probability, and replace $\cap$ with "and":

$P(A \cup B) = P(A) + P(B) - P(A \cap B)$

**Example** (Devore Chapter 2, problem 12)

Consider randomly selecting a student at a certain university, and let $A$ denote the event that the selected individual has a Visa credit card and $B$ be the analogous event for a MasterCard. Suppose that $P(A) = 0.5$, $P(B) = 0.4$, and $P(A \cap B) = 0.25$.

1. Compute the probability that the selected individual has at least one of the two types of card (i.e., the probability of the event $A \cup B$).
2. What is the probability that the selected individual has neither type of card?
3. Describe, in terms of $A$ and $B$, the event that the selected student has a Visa card but not a MasterCard, and then compute the probability of this event.

Provide both visual "proofs" or mathematical calculations.

▼ **Answers:**

1. $P(A \cup B) = P(A) + P(B) - P(A \cap B) = 0.5 + 0.4 - 0.25 = 0.65$
2. $P(A' \cap B') = 1 - P(A \cup B) = 0.35$ (Based on visual derivation)
3. $P(A \cap B') = P(A) - P(A \cap B) = 0.5 - 0.25 = 0.25$ (Based on visual derivation)

Typically, we don't do mathematical proofs on sets – demonstrations with Venn diagrams are usually sufficient.

## 1.19 DeMorgan's Laws

(Not covered)

$(A \cup B)' = A' \cap B'$

$(A \cap B)' = A' \cup B'$

Proofs

## 1.20 Conditional Probability and the Multiplication Rule

We want to know the probability of event $A$ given event $B$ occurred. One way to do this is to count and write down how we expect the number of times $A$ occurred, given that event $B$ occurred, in terms of set operations. First, consider the fraction

$$F(A \text{ given } B) = \frac{n(A \cap B)}{n(A \cap B) + n(A' \cap B)}$$

The numerator is the number of times $A$ and $B$ occurred.

The denominator is the number of times $B$ occurred – it can occur when $A$ did or did not occur.

Note that $A \cap B$ and $A' \cap B$ are mutually exclusive, so $(A \cap B) \cup (A' \cap B) = B$, so we can also write

$$F(A \text{ given } B) = \frac{n(A \cap B)}{n(B)}$$

which is also visually obvious from a diagram.

Dividing all terms on the right-hand side by $n$, using the definition of probability in terms of relative frequency and introducing the symbol "|" gives the definition of conditional probability, provided $P(B) > 0$:

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

We were given that the probability of a student having a Visa is $0.5$; the probability of a student having a MasterCard is $0.4$; and the probability of a student having both is $0.25$.

We were asked to find the probability that the student has a Visa but not a MasterCard.

How is this different from the statement "given the student has a Visa, what is the probability that they do not have a MasterCard?"

▼ **Answer**

In the first case, we don't know anything about any of the students. In the second case, we are told to only consider a subset of all students. Our new sample space contains only the $A$ (Visa) part of the original sample space.

"Has a Visa but not a MasterCard" is the event $V \cap M'$, with probability $0.25$. In contrast, "does not have a MasterCard, given that the student has a Visa" has conditional probability $P(M'|V) = 0.25/0.5 = 0.5$.

Using

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

we need to compute the right-hand side of

$$P(M'|V) = \frac{P(M' \cap V)}{P(V)}$$

Based on the Venn diagram, we know $P(M' \cap V) = 0.25$ and we were given $P(V) = 0.5$, so

$$P(M'|V) = \frac{1}{2}$$

$P(A \cap B) = P(A|B)P(B)$ is sometimes called the multiplication rule

**Example**

If you are in a firing line and two people have guns that shoot a real bullet instead of a blank with probability of 1/3, what is the probability that you get shot (assuming the marksmen never miss)?

*Answer*

Assume the two live/blank outcomes are independent.

$P(A \text{ or } B) = P(A) + P(B) - P(A \cap B)$

$P(A \cap B) = P(A|B)P(B)$

If $A$ and $B$ are independent, $P(A|B) = P(A)$, so

$P(A \cap B) = P(A)P(B)$. Thus

$P(A \text{ or } B) = 1/3 + 1/3 - (1/3)(1/3) = 5/9$

or, $P(\overline{A \text{ or } B}) = (2/3)(2/3) = 4/9$

Check: $P(\overline{A \text{ or } B}) = 1 - P(A \text{ or } B)$

## 1.21 Law of Total Probability

Let $A_1, \ldots, A_k$ be mutually exclusive and exhaustive events. Then for any other event $B$,
$P(B) = P(B|A_1)P(A_1) + ... + P(B|A_k)P(A_k)$

Explain this using a table and a Venn diagram.

Consider a square partitioned by three non-overlapping rectangles. Draw $B$ as a rectangle inside the square. We can count the number of elements in $B$ using conditional counts:

$n(B) = n(B|A_1) + n(B|A_2) + n(B|A_3)$

Using $P(B|A_1) = n(B|A_1)/n(A_1)$, etc., we have

$n(B) = P(B|A_1)n(A_1) + P(B|A_2)n(A_2) + P(B|A_3)n(A_3)$

Divide both sides by $n$ to arrive at the result.

## 1.22 Bayes' Rule

$$P(A|B) = P(A) \cdot \frac{P(B|A)}{P(B)}$$

Also called "Bayes' Law" and "Bayes' Theorem".

Further reading: The Prosecutor's Fallacy; The Sally Clark Case; Bayes' theorem, COVID19, and screening tests; Fooling Juries

**Simple Derivation**

Definition of conditional probability for two events:

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

Swapping letters gives

$$P(B|A) = \frac{P(B \cap A)}{P(A)}$$

The numerators are identical because $A \cap B = B \cap A$. Combining these two equations gives Bayes' rule.

▼ **Visual Derivation/Exploration**

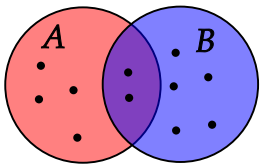



**Figure 1**

**Question:** What is $n(A \text{ and } B)$ in Figure 1? That is, how many dots have labels of both $A$ and $B$? (Give a number)

**Answer**: 2

**Question:** What is $n(B \text{ and } A)$ in Figure 1? That is, how many dots have labels of both $A$ and $B$? (Give a number)

**Answer**: 2

---

**Question:** In terms of $n(A), n(B), n(A \text{ and } B)$, what is $n(A \text{ or } B)$ in Figure 1? That is, how many dots have labels of $A$ or $B$?

**Answer**: $n(A \text{ or } B) = n(A) + n(B) - n(A \text{ and } B)$

**Question:** In terms of $n(B)$ and $P(A|B)$, what is $n(A \text{ and } B)$ in Figure 1?

$n(A \text{ and } B) = n(B) \cdot P(A|B)$

**Question:** In terms of $n(A)$ and $P(B|A)$, what is $n(B \text{ and } A)$ in Figure 1?

$n(B \text{ and } A) = n(A) \cdot P(B|A)$

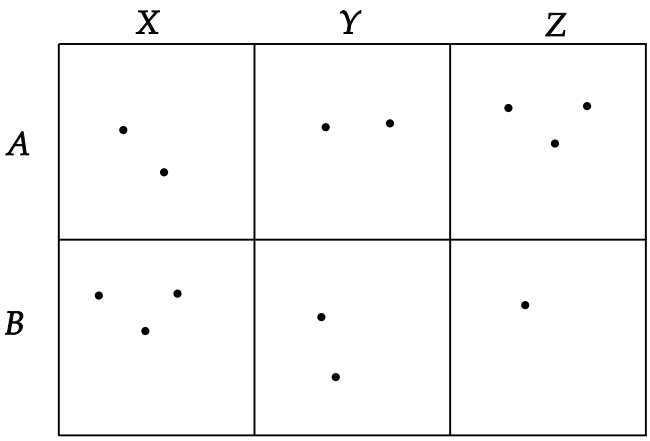



**Figure 2**

**Question:** In terms of $n(A), n(B), n(X), P(X|A)$, and $P(A|X)$, what is $n(A \text{ and } X)$ in Figure 2?

$n(A \text{ and } X) = n(A) \cdot P(X|A)$

**Question:** In terms of $n(A), n(B), n(X), P(X|A)$, and $P(A|X)$, what is $n(X \text{ and } A)$ in Figure 2?

**Answer:** $n(X \text{ and } A) = n(X) \cdot P(A|X)$

**Question:** In reference to Figure 2, what is $P(A|B)$ in terms of $n(A), n(B), n(A \text{ and } B)$, and $n(B \text{ and } A)$?

$$P(A|B) = \frac{n(A \text{ and } B)}{n(B)}$$

**Question**: In reference to Figure 2, what is

$$\frac{n(A \text{ and } B)}{n(B \text{ and } A)}$$

in terms of $n(A), n(B), P(A|B)$, and $P(B|A)$?

**Answer:**

$$\frac{n(A \text{ and } B)}{n(B \text{ and } A)} = 1 = \frac{n(A) \cdot P(B|A)}{n(B) \cdot P(A|B)}$$

or

$$n(B) = n(A) \cdot \frac{P(B|A)}{P(A|B)}$$

or

$$P(B) = P(A) \cdot \frac{P(B|A)}{P(A|B)}$$

**Question:** In reference to Figure 2, what is

$$\frac{n(A \text{ and } X)}{n(X \text{ and } A)}$$

in terms of $n(A), n(X), P(A|X)$, and $P(X|A)$?

**Answer**

$$\frac{n(A \text{ and } X)}{n(X \text{ and } A)} = 1 = \frac{n(A) \cdot P(X|A)}{n(X) \cdot P(A|X)}$$

or

$$n(X) = n(A) \cdot \frac{P(X|A)}{P(A|X)}$$

or

$$P(X) = P(A) \cdot \frac{P(X|A)}{P(A|X)}$$

check

$$\frac{5}{13} = \frac{7}{13} \cdot \frac{\frac{2}{7}}{\frac{2}{5}}$$

### 1.22.1 Example

A cab was involved in a hit-and-run accident at night. Two cab companies, the Green and the Blue, operate in the city. You are given the following data:

- 85% of the cabs in the city are Green, and 15% are Blue. A witness identified the cab as Blue. The court tested the reliability of the witness under the circumstances that existed on the night of the accident and concluded that the witness correctly identified each one of the two colors 80% of the time and failed 20% of the time.

What is the probability that the cab involved in the accident was Blue rather than Green? Use the two approaches (equation- and diagram- based).

**Answer**

**Method 1**

Consider 1000 recreations of the incident in which 850 vehicles are Green, and 150 vehicles are Blue. Based on a correct identification of 80% the expected number for each possible witness claim is shown in the last column.

```
                        850*0.80 = 680 - Is Green, claims Green                 1
       850 Are Green                                                            2
                        850*0.20 = 170 - Is Green, claims Blue                  3
1000                                                                            4
                        150*0.80 = 120 - Is Blue, claims Blue                   5
       150 Are Blue                                                             6
                        150*0.20 = 30  - Is Blue, claims Green                  7
```

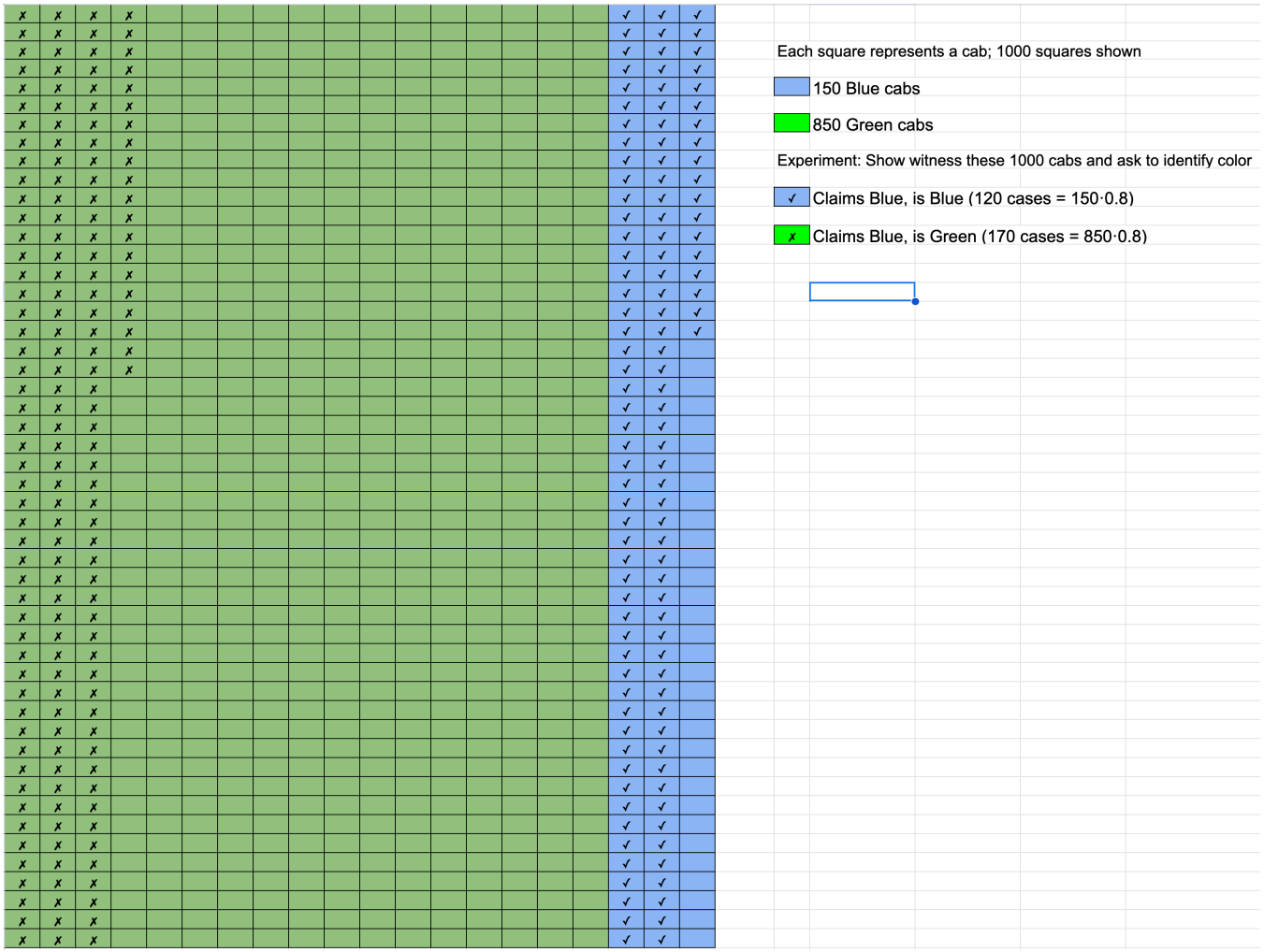
We want to know the probability the cab is Blue when the witness claimed Blue. The number of times in the last column where the witness claimed Blue is $170 + 120$ (middle two rows). The number of times this claim is correct is $120$.

So the probability the cab is Blue given the witness claimed Blue is

$$P(B|W_B) = \frac{120}{120 + 170} \approx 0.41$$

**Method 2**

The following is an alternative visualization of the tree diagram of **Method 1**.




**Method 3**

To use Bayes' theorem, we start by writing the given probabilities

- $P(G) = 0.85$ (Probability a cab is Green)
- $P(B) = 0.15$ (Probability a cab is Blue)
- $P(W_B|B) = 0.80$ (Probability witness claims Blue when Blue)
- $P(W_B|G) = 0.20$ (Probability witness claims Blue when Green)

$$P(B|W_B) = P(W_B|B)\frac{P(B)}{P(W_B)}$$

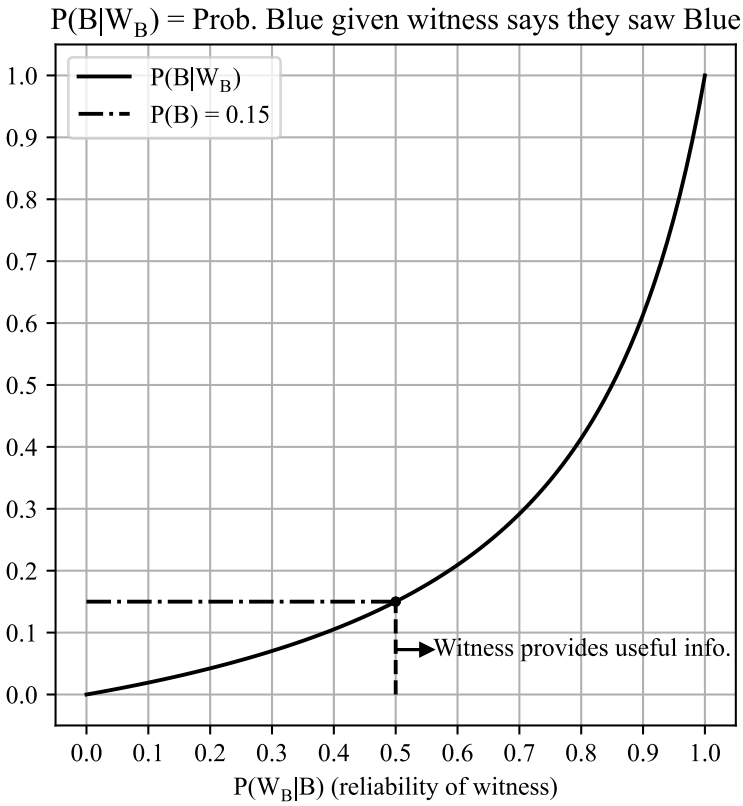
The denominator is
$P(W_B) = P(B)P(W_B|B) + P(G)P(W_B|G) = 0.15 \cdot 0.80 + 0.85 \cdot 0.20 = 0.12 + 0.17$. Thus,

$$P(B|W_B) = 0.80\frac{0.15}{0.15 \cdot 0.80 + 0.85 \cdot 0.20} = \frac{0.12}{0.12 + 0.17}$$

Multiplying the numerator and the denominator by 1000 gives the same equation for **Method 1**.

$$P(B|W_B) = \frac{120}{120 + 170} \approx 0.41$$

A plot of $P(B|W_B)$ vs witness accuracy ("reliability") is given below. If the witness is less than 50% reliable, $P(B|W_B)$ is less than $P(B)$. It would seem that this witnesses testimony should be ignored. However, suppose the witness has a reliability of zero. In this case we know that when the witness says "blue" the car was actually green! However, I doubt the witness would be allowed to testify if they were less than 50% reliable. (The label in the plot should be "witness information gives same result as $P(B)$".).




## 1.22.2 Terminology

(Not covered yet – will need in future.)

$$P(A|B) = P(B|A)\frac{P(A)}{P(B)}$$

- Posterior: $P(A|B)$ (probability after knowing $B$ occurred)
- Prior: $P(A)$ (probability prior to knowing $B$ occurred)
- Marginal probability: $P(B)$ (why "marginal"?)
- Likelihood: conditional probability on right–hand side, $P(B|A)$
- Prior odds for $A$ versus $A'$: $P(A)/P(A')$

Other forms of Bayes include

$$\text{posterior probability} = \text{likelihood} \cdot (\text{prior probability})/(\text{marginal probability})$$

and the proportionality

$$\text{posterior} \propto \text{likelihood} \cdot \text{prior}.$$

See also Understanding Bayes Theorem with Ratios, which uses

$$\text{original odds} \cdot \text{likelihood ratio} = \text{new odds}$$

▼ **More Terminology**

In medical terminology (see also Wikipedia; notes by ekamperi; and Covid examples: 1 | 2 | 3),

- Sensitivity, $S_e$ (true positive rate):

  $P(T^+|D^+)$ = (number of true positives)/(n true positives + n false negatives)

  $P(T^+|D^+)$ = (true positives)/(total number with disease)

  where $T^+$ is a positive test result and $D^+$ means "disease present"

- Specificity, $S_p$ (true negative rate):

  $P(T^-|D^-)$ = (number of true negatives)/(number of true negatives + number of false positives)

  $P(T^-|D^-)$ = (number of true negatives)/(total number without disease).

  where $T^-$ is a negative test result and $D^-$ means "disease absent"

- Likelihood ratio: (See also The likelihood ratio and its graphical representation):
  $LR(r) = P(r|D^+)/P(r|D^-)$, where $r$ is the test result. For a continuous variable such as "HDL cholesterol", use the ratio of conditional probability densities rather than point probabilities. Then

  Post-test odds of $D^+$ = $LR(r)$ · Pre-test odds of $D^+$

  If $r$ is dichotomous (test result is positive or negative), then

  $LR^+ = P(T^+|D^+)/P(T^+|D^-) = S_e/(1 - S_p)$

  and

  $LR^- = P(T^-|D^+)/P(T^-|D^-) = (1 - S_e)/S_p$

### 1.23 General Bayes' Rule

When $A_1, \ldots, A_k$ are mutually exclusive and exhaustive and $P(B) > 0$, we can write

$$P(A_j|B) = \frac{P(B|A_j)P(A_j)}{P(B|A_1)P(A_1) + ... + P(B|A_k)P(A_k)}$$

## 2 Counting

Three types of problems:

1. Product Rule:

   A. Given $k$ ordered boxes and $n_1$ choices for first box, $n_2$ for the second, …

   B. Given $k$ ordered boxes and $n$ choices for first box, $n$ for second, …

2. Permutations: Given **one** set of size $n$, how many distinct *ordered* $k$-tuples with no duplicate elements can be created? (e.g., set = {a, b}, permutations of length 2 are $(a, b)$ and $(b, a)$). Similar to a product rule B. problem where $n_1 = n, n_2 = n - 1, \ldots$ .

3. Combinations: Same as 2. except counting all sets with the same elements as equivalent. (e.g., if set = {a, b} only one combination is possible: {a, b}).

▼ **Stirling's Approximation**

You will often encounter $N!$ in counting problems. It is useful to know Stirling's approximation to estimate $N!$ for large $N$:

$\ln N! \simeq N \ln N - N$

or

$\ln N! \simeq N \ln N - N + \ln \sqrt{2\pi N}$

From this form, it follows that

$N! \simeq N^N e^{-N} \sqrt{2\pi N}$

I recommend remembering the first form $\ln N! \simeq N \ln N - N$ for a rough approximation, but note that the factorial approximation including $\sqrt{2\pi N}$ has a relative error that approaches zero as $N$ increases. See stirling.py.

$N = 10$

$N! \simeq 3.63 \cdot 10^6$

$e^{N \ln(N) - N} \simeq 4.54 \cdot 10^5$

$e^{N \ln(N) - N + \ln \sqrt{2\pi N}} \simeq 3.60 \cdot 10^6$

$N = 100$

$N! \simeq 9.33 \cdot 10^{157}$

$e^{N \ln(N) - N} \simeq 3.72 \cdot 10^{156}$

$e^{N \ln(N) - N + \ln \sqrt{2\pi N}} \simeq 9.32 \cdot 10^{157}$

### 2.0.1 Product Rule (or Law of Multiplication)

(Devore does not name it but gives it as a proposition on p 65)

> If the first element or object of an ordered pair can be selected in $n_1$ ways, and for each of these $n_1$ ways the second element of the pair can be selected in $n_2$ ways, then the number of pairs is $n_1 n_2$.

One can use a tree diagram, table, or $x$–$y$ plot to justify.

**Tree Diagram**

Use for visually justifying the product rule and counting permutations (Devore p 66)

**Tuple**

A "$k$–tuple" is an ordered collection of $k$ objects. (Devore p 66)

***General Product Rule***

AKA Product Rule for $k$-Tuples

Suppose a set consists of ordered collections of $k$ elements ($k$-tuples) and that there are $n_1$ possible choices for the first element; for each choice of the first element, there are $n_2$ possible choices of the second element; …; for each possible choice of the first $k-1$ elements, there are $n_k$ choices of the $k$th element. Then there are $n_1 n_2 ... n_k$ possible $k$-tuples. (Devore p 66)

Note that "elements" is used here, but in the definition of a tuple, objects is used.

**Example**

Take two steps; each step is North, South, East, or West.

Put one of N, S, E, W in the first box and the same for the second box. Result is 16 unique step pairs.

Tree diagram.

Equivalent problem: Sample with replacement from set {N, S, E, W}.

**Example**:

If operation 1 is moving north, south, east, or west and operation 2 is moving up or down, then there are 8 possible operations of length 2.

**Example**:

Two teams of twelve players each. How many unique handshakes between members of opposing teams?

Use a tree diagram for teams of 3 first, then generalize.

▼ **Answer**

$n_a = 12, n_b = 12, N = 12 \cdot 12 = 144$.

**Example**: Roll a die five times. How many 5-tuples?

▼ **Answer**

Create five boxes. There are six possible "choices" for the first box, six possible choices for the second box, ….. So there are $6^5$ possible 5–tuples.

**Example**: Flip a coin 2 times.

▼ **Answer**

The number of 2–tuples is $2 \cdot 2$. (Think of two boxes and you put either a $H$ or $T$ in the first box and a $H$ or $T$ in the second box.)

**Example**: Each clinic has two $O$ doctors and three $P$ doctors, and you must select one doctor of each type from the same clinic. How many possible pairs of $O$s and $P$s are there per clinic?

▼ **Answer**

In the first box, put one of the two $O$s. For each $O$, there are 3 $P$s to choose from and put in the second box. So $n = 2 \cdot 3$.

If each clinic also has three $I$s and two $G$s, how many possible choices are there per clinic for four doctors, one of each type?

▼ **Answer**

In the third box, put one of the three $I$s; in the fourth box, put one of the two $G$s. Then $n = 2 \cdot 3 \cdot 3 \cdot 2$.

**Example**: Suppose you want to pick a team of two tennis players from 3 players, $A$, $B$, and $C$.

▼ **Answer**

The number of ordered selections is $3 \cdot 2$: $AB$, $AC$, $BA$, $BC$, $CA$, and $CB$.

This is not the list of distinct teams because $AB$ is the same as $BA$ (that is, order is not important). The number of possible teams is 3, by inspection.

### 2.0.2 Permutation

An ordered arrangement of distinct objects, where each arrangement has no duplicate objects. Usually relevant in problems that involve "without replacement".

**Example**:

You have stickers labeled $1, \ldots, 6$ that are used to form a license plate.

How many unique license plates of length 4 can you form?

▼ **Answer**

$6 \cdot 5 \cdot 4 \cdot 3 = 360$

To see relationship to $P_{k,n}$ formula given next, consider

$$6 \cdot 5 \cdot 4 \cdot 3 = \frac{6!}{(6-4)!} = \frac{6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1}{2 \cdot 1}$$

Suppose you have $n$ distinct objects and you want to put them in boxes labeled $1, 2, \ldots, k$. You select one object and put it in the first box. You select a second object from the remaining $n-1$ objects and put it in box 2, ….

The number of ways to do this is denoted $P_{k,n}$ (or ${}_nP_k$) and is

$$P_{k,n} = \frac{n!}{(n-k)!} = n \cdot (n-1) ... \cdot (n-k+1) = \frac{n \cdot (n-1) \cdot (n-2) ... \cdot (n-k) \cdot (n-k-1) ... 1}{(n-k) \cdot (n-k-1) ... 1}$$

**Example**

Step {N, S, E, W}. Then take another step, but not in the same direction as first.

▼ **Answer**

$4 \cdot 3 = 12$

**Example**

A four-volume work is placed in random order on a bookshelf. What is the probability of the volumes being in proper order $(1, 2, 3, 4)$?

▼ **Answer**

$1/4!$

**Example**

A subway train made up of $n$ cars is boarded by $r$ passengers ($r \leq n$), each entering a car completely at random.

1. What is the number of ways the passengers can board?
2. What is the probability of the passengers all ending up in different cars?

▼ **Answer**

1. Consider list of $r$ passengers and each can be assigned number $1, ...n$: $n^r$
2. Have $n$ choices for the first passenger, $n-1$ for the second, … $n-r+1$ for the last: $\frac{n!}{(n-r)!, n^r}$.

## 2.0.3 Combination (un-ordered subset)

The number of unique $k$–tuples if $k$–tuples with the same elements (but in a different order) are treated as the same. In the tennis team-picking example, there are 3 possible team combinations.

Each permutation can be regarded as a group of $k$. If we regard groups as equivalent if they have the same elements, then there are fewer groups than permutations. For example, if the two permutations

$(1, 2)$

$(2, 1)$

are regarded as equivalent, then there is only one group containing the numbers 1 and 2. To determine the number of possible orderings of each permutation, ask how many ways a set of $k$ elements can be arranged. The answer is $k!$.

So, to find the number of combinations, divide the number of permutations by $k!$.

$$C_{n,k} = \frac{P_{k,n}}{k!} = \frac{\frac{n!}{(n-k)!}}{k!} = \frac{n!}{k!(n-k)!}$$

$C_{n,k}$ is often called a binomial coefficient, denoted by $\binom{n}{k}$, and referred to as "$n$ choose $k$".

**Example**

Select two players from a list of three.

1. Assign one as captain. How many unique teams?
2. If there is no assignment of a captain, how many unique teams?

**Example**:

How many unique ordered hands of size 5 can be formed using a 52-card deck?

▼

This is a permutation problem: $52 \cdot 51 \cdot 50 \cdot 49 \cdot 48$ permutations.

**Example**:

How many hands of size 5 can be formed using a 52-card deck?

▼

Each permutation can be rearranged in 5! ways. So the number of hands (combinations) is $52 \cdot 51 \cdot 50 \cdot 49 \cdot 48/(5 \cdot 4 \cdot 3 \cdot 2 \cdot 1)$

# 3 Random Variables and Distributions

For a given sample space $\mathcal{S}$ of some experiment, a random variable (rv) is any rule that associates a number with each outcome in $\mathcal{S}$. In mathematical language, a random variable is a function whose domain is the sample space and whose range is the set of real numbers. (Devore p 93)

Example: $\mathcal{S} = \{T, F\}$ with $X(T) = 1$ and $X(F) = 0$ defines the discrete random variable $X$ that maps outcomes in $\mathcal{S}$ to a number.

Example: If an experiment is to flip a coin until an $H$ is encountered, $\mathcal{S} = \{H, TH, TTH, ...\}$ and $X(H) = 1, X(TH) = 2, X(TTH) = 3$ defines the random variable $X$ as the number of flips until an $H$ is encountered.

Any random variable whose only possible values are 0 and 1 is called a Bernoulli random variable. (Devore p 94)

## 3.1 Discrete

### 3.1.1 Random Variables

A **discrete** random variable is an rv whose possible values either constitute a finite set or else can be listed in an infinite sequence in which there is a first element, a second element, and so on ("countably" infinite).

A random variable is **continuous** if both of the following apply:

1. Its set of possible values consists either of all numbers in a single interval on the number line (possibly infinite in extent, e.g., from $-\infty$ to $\infty$) or all numbers in a disjoint union of such intervals (e.g., $[0, 10] \cup [20, 30]$).
2. No possible value of the variable has positive probability, that is, $P(X = c) = 0$ for any possible value $c$.

(Devore p 95)

A countably infinite set means one can match each element in the set to a natural number $(0, 1, 2, \ldots)$. A non-countable set is the real numbers in the interval $[0, 1]$, as proved by Cantor.

### 3.1.2 PMF Definition

The probability distribution or probability mass function (pmf) of a discrete rv is defined for every number $x$ by $p(x) = P(X = x) = P(\text{all } s \in \mathcal{S} : X(s) = x)$.

$p(x) \geq 0$ and $\sum_x p(x) = 1$ are required for any pmf.

A cumulative distribution function (cdf) is the running sum of the pmf. It is denoted by $F(x) = P(X \leq x)$. Its interpretation is the probability that $X$ will be at most $x$.

$$P(X \leq x) = \sum_{y \leq x} p(y)$$

Example: $P(X = 1) = 0.2$, $P(X = 2) = 0.3$, $P(X = 3) = 0.5$ defines a pmf for random variable $X$ that can take on values $1$, $2$, and $3$. It has a cdf of

$P(X \leq 1) = 0.2$

$P(X \leq 2) = 0.2 + 0.3 = 0.5$

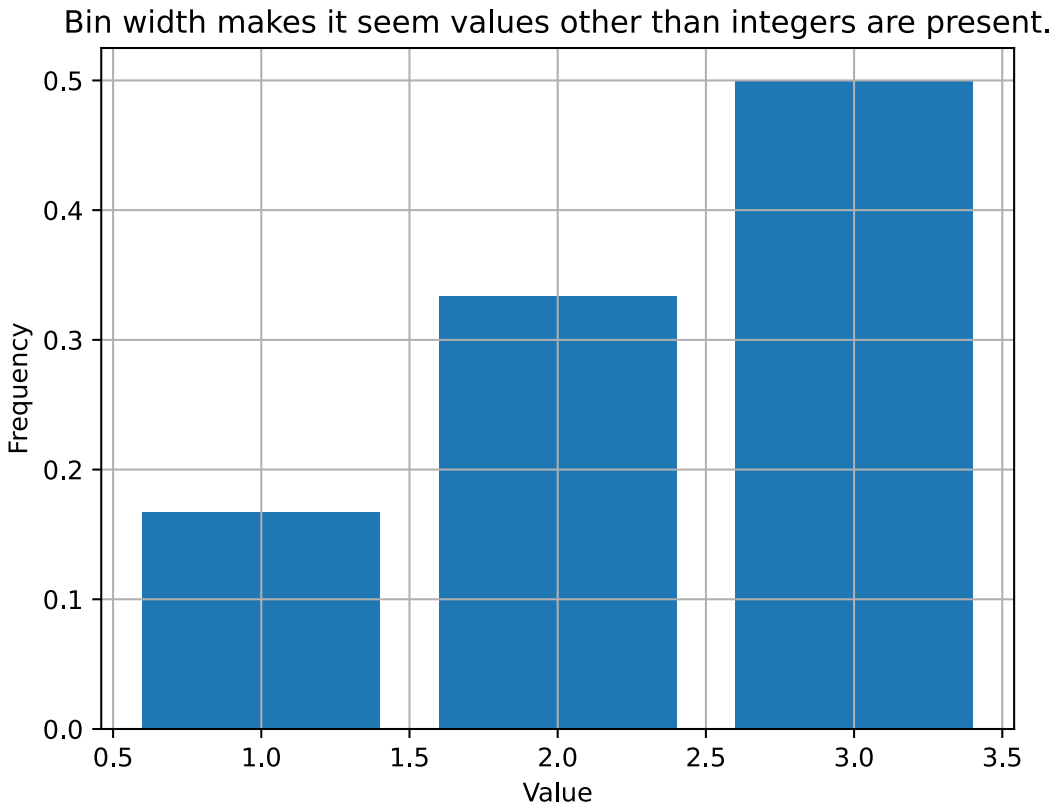
$P(X \leq 3) = 0.2 + 0.3 + 0.5 = 1$

### 3.1.3 Plotting

Code for generating these plots: pmf.py.

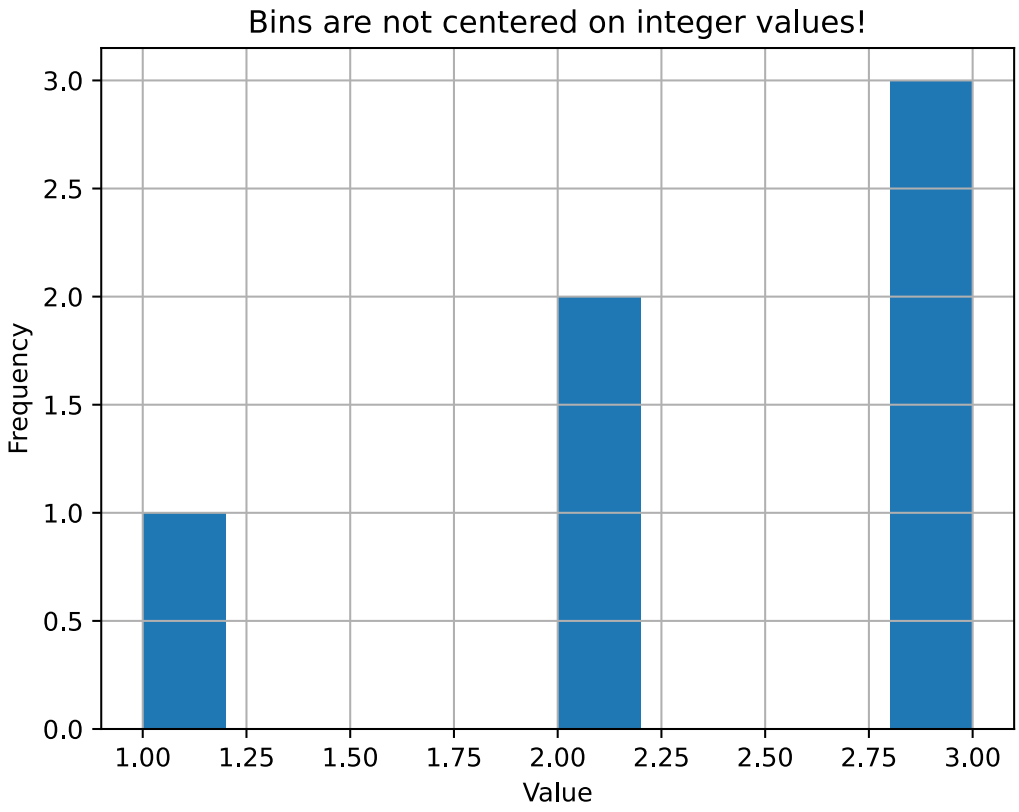
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Bad example. Bin width is misleading.

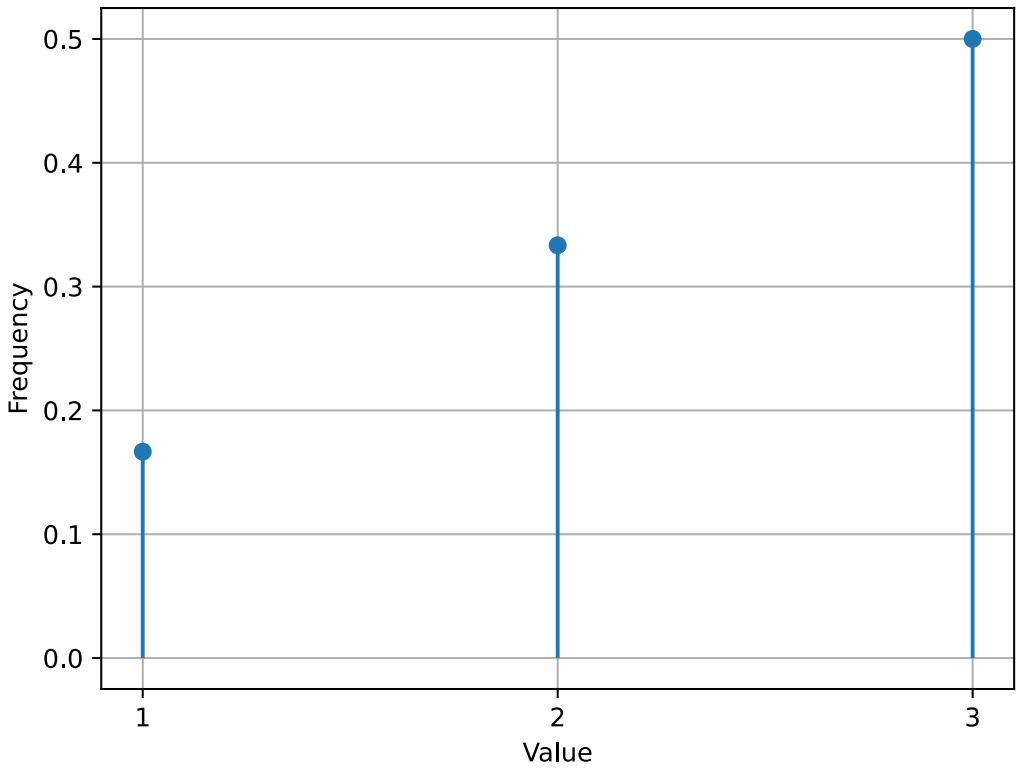



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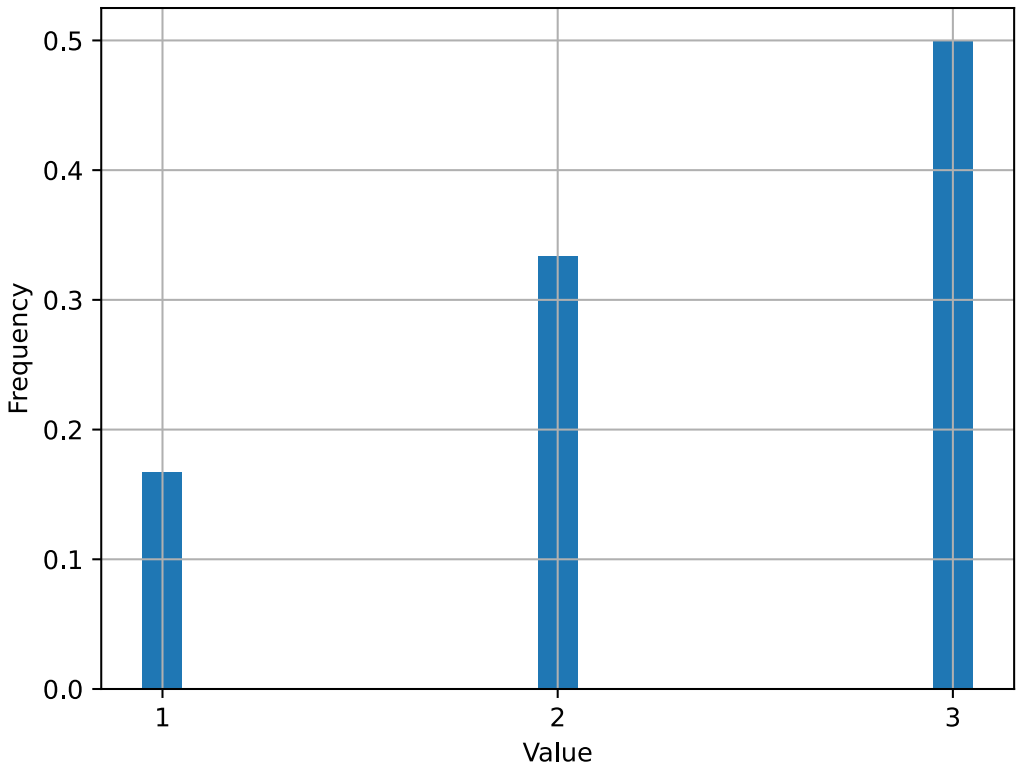
Bad example. Bins are not centered on integers.



PMF showing relative frequency.

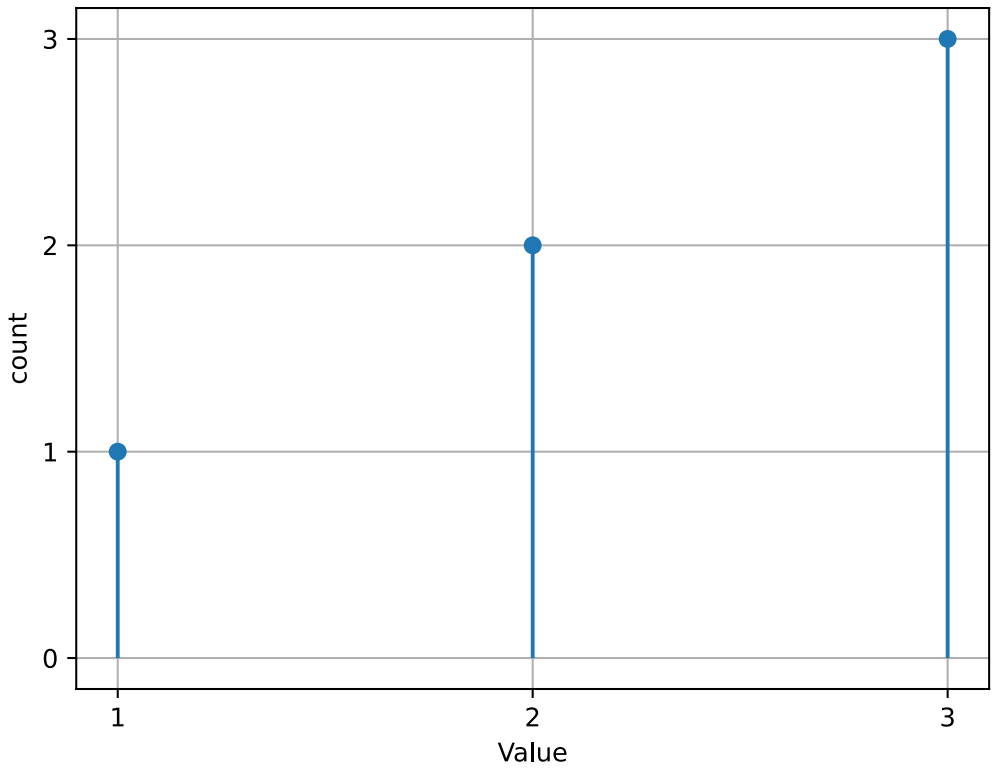


Same as above, but using thin bars.

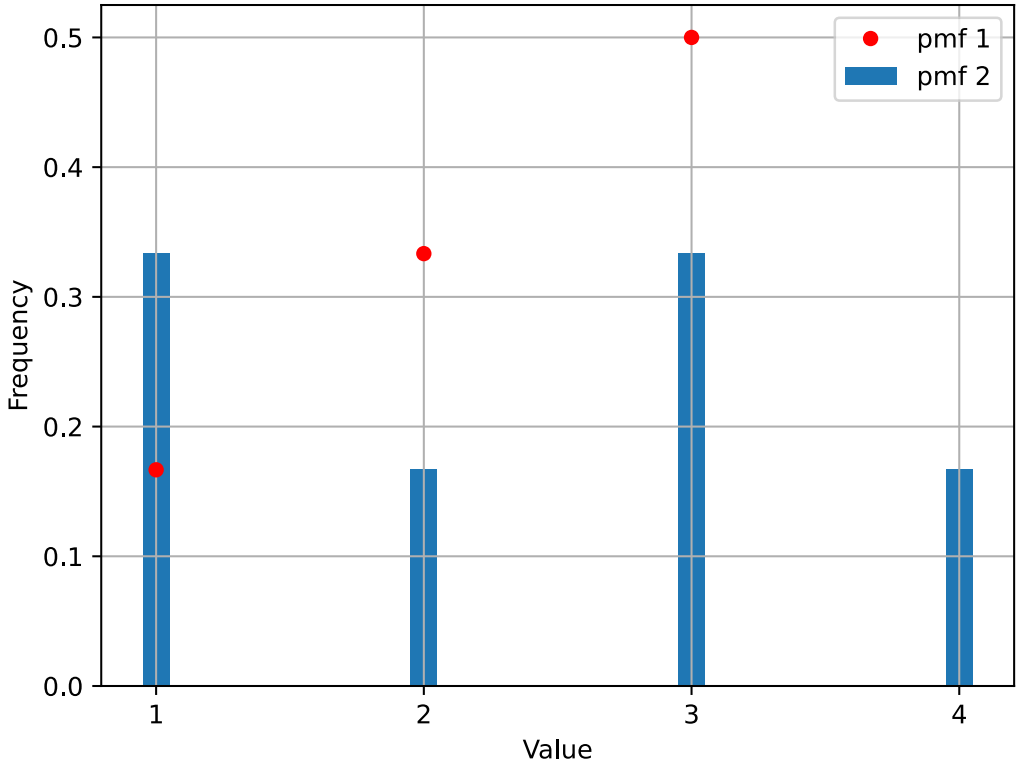



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Histogram showing counts (not a normalized PMF).



An option for plotting two PMFs




### 3.1.4 Expectation Values

$E(h(X))$ or $E[h(X)]$ is the notation.

> If the random variable $X$ has a set of possible values $D$ and pmf $p(x)$, then the expected value of any function $h(X)$, denoted by $E[h(X)]$ or $\mu_{h(X)}$, is computed by
>
> $E[h(X)] = \sum_{D} h(x)p(x)$
>
> (Devore p 109)

Example:

$E[X^2]$ where $X$ is the random variable with pmf of $P(X=1) = 0.2, P(X=2) = 0.3, P(X=3) = 0.5$ is

$E[X^2] = 1^2 \cdot 0.2 + 2^2 \cdot 0.3 + 3^2 \cdot 0.5 = 5.9$

#### 3.1.4.1 Mean

$h(x) = x$ and we define $\mu$ according to $E[X] = \mu$

#### 3.1.4.2 Variance

$h(x) = (x - \mu)^2$ and we define $V$ and $\sigma_X$ according to $V(X) = \sigma_X^2 = E[(X - \mu)^2]$.

We also define the standard deviation as $\sigma_X = \sqrt{\sigma_X^2}$.

It can be shown that

$V(X) = E[(X - \mu)^2] = E[X^2] - (E[X])^2$

or equivalently

$V(X) = E[X^2] - \mu^2 = \sum_D x^2 p(x) - \mu^2$

We will use this result later when we consider estimating $V$ by taking random samples from a population (instead of using the full population).

▼ **Justification**

$E[(X - \mu)^2] = E[X^2 - 2\mu X + \mu^2] = E[X^2] - E[2\mu X] + E[\mu^2]$

Using

$E[2\mu X] = 2\mu E[X] = 2\mu^2$ gives

$E[(X - \mu)^2] = E[X^2] - \mu^2$

## 3.1.5 Important PMFs

There are several key probability mass functions. For each distribution, we want to know properties such as its mean and variance.

### 3.1.5.1 Binomial

There are many experiments that conform either exactly or approximately to the following list of requirements:

1. The experiment consists of a sequence of $n$ smaller experiments called trials, where $n$ is fixed in advance of the experiment.
2. Each trial can result in one of the same two possible outcomes (dichotomous trials), which we generically denote by success (S) and failure (F).
3. The trials are independent, so that the outcome on any particular trial does not influence the outcome on any other trial.
4. The probability of success $P(S)$ is constant from trial to trial; we denote this probability by $p$. An experiment for which Conditions 1-4 are satisfied is called a binomial experiment. (Devore p 114)

The binomial random variable $X$ associated with a binomial experiment consisting of $n$ trials is defined as

$X$ = the number of $S$'s among the $n$ trials

(Devore p 114)

The pmf of a binomial random variable is

$$b(x;n,p) = \binom{n}{x} p^x (1-p)^{n-x}$$

Notation: for any pmf that has a name, we sometimes write, e.g.,

$X \sim b(x;n,p)$

To mean the random variable $X$ has a pmf given by $b$.

### 3.1.5.2 Derivation of Binomial Coefficients

General problem: Given $n$ positions, how many arrangements with $x$ objects of one type and $n-x$ of another are possible? The number is $C_{n,x}$.

Use the label $p$ for the $x$ objects and $q$ for the $n-x$ objects. Suppose $n=3$. All possible permutations are listed. The ones that satisfy $x=2$ are indicated with a $*$. Thus, $C_{3,2}=3$.

```
ppp                1
ppq  *             2
pqp  *             3
pqq                4
qpp  *             5
qpq                6
qqp                7
qqq                8
```

Also, by inspection, $C_{3,0}=1$, $C_{3,1}=3$, and $C_{3,3}=1$. In summary

$C_{3,0}=1, C_{3,1}=3, C_{3,2}=3, C_{3,3}=1$

We want a general formula for computing $C_{n,x}$.

**Method I**

A mathematical shortcut for finding $C_{n,x}$ is to note that the above table can be generated using $(p+q)^3$, which has 8 terms when expanded, corresponding to the rows in the table. However, it simplifies to

$(p+q)^3 = p^3 + 3p^2q + 3pq^2 + q^3$

The coefficients in the simplified form count the arrangements with each possible value of $x$, which is what we want.

(Note that the coefficients $1,3,3,1$ are in row $n=3$ of Pascal's triangle, counting the top row as $n=0$. See also Larson p 46 for a derivation of the rule that relates rows in Pascal's triangle.)

We happen to have an equation that gives us the simplified form. The binomial theorem is

$$(p+q)^n = \sum_{x=0}^{n} \binom{n}{x} p^x q^{n-x}$$

where

$$\binom{n}{x} = \frac{n!}{x!(n-x)!}$$

Therefore, we conclude

$$C_{n,x} = \binom{n}{x}$$

**Method II**

See Bulmer, Chapter 6 and Chapter 1 of Kittel and Kroemer.

General problem: In $n$ independent trials with success probability $p$, what is the probability of exactly $x$ successes?

Recall the table

| | |
|---|---|
| ppp | 1 |
| ppq | 2 |
| pqp | 3 |
| pqq | 4 |
| qpp | 5 |
| qpq | 6 |
| qqp | 7 |
| qqq | 8 |

If we regard $p$ as a probability and define $q = 1 - p$, then the probability of each row is obtained by multiplication. Some rows have the same probability. For example, $pqq = qpq = qqp$. Based on this, we can conclude that

$C_{3,0}$ has probability $p^3$

$C_{3,1}$ has probability $3pq^2 = 3p(1-p)^2$

$C_{3,2}$ has probability $3p^2q = 3p^2(1-p)$

$C_{3,3}$ has probability $q^3 = (1-p)^3$

or

$$P(X = x) = \binom{n}{x} p^x (1-p)^{n-x}$$

We write the probability in a more specific form as

$$b(x;n,p) = \binom{n}{x} p^x (1-p)^{n-x}$$

where the values after the semicolon are constants.

#### 3.1.5.3 Mean and Variance

Derivation

#### 3.1.5.4 Hypergeometric

#### 3.1.5.5 Negative Binomial

#### 3.1.5.6 Poisson Distribution and Poisson Process

References

- The original Poisson paper is in French but is covered in English by Stigler 1982. Poisson used the limit of the binomial distribution, and Stigler notes that De Moivre derived a related approximation to the Poisson formula.
- How used in physics lab experiments using Geiger counters: 1, 2, 3
- Derived independently by Bateman using a differential equation approach in Rutherford, Geiger, and Bateman, 1910
- A simple derivation from limit of binomial is given in lecture notes by D.S.G. Pollock. Key steps:
  1. Assume $\mu = np$ is constant and $n \to \infty$ (so $p$ becomes small as $n$ becomes large; the probability of an event becomes vanishingly small).
  2. Let $p = \mu/n$ an substitute into $\frac{n!}{(n-x)!x!} p^x (1-p)^{n-x}$
  3. Show that

     $$\lim_{n\to\infty} \frac{n!}{(n-x)!n^x} = 1$$

  4. Show that

     $$\lim_{n\to\infty} \left(1 - \frac{\mu}{n}\right)^{-x} = 1$$

  5. Note that

     $$\lim_{n\to\infty} \left(1 - \frac{\mu}{n}\right)^{n} = e^{-\mu}$$

The Poisson probability mass function is

$$P(x) = \frac{\mu^x e^{-\mu}}{x!}$$

The symbol $\mu$ was chosen because $E[X] = \mu$ for the Poisson. This distribution can be derived from the Binomial distribution in the limit $n \to \infty$ with $x$ and $\mu \equiv np$ fixed. That is, we let $n$ become large at the same rate that $p$ becomes small. So this corresponds to a Binomial experiment where the probability of a success is small.

The Poisson process is a process that satisfies

1. in a sufficiently short amount of time, $\Delta t$, only 0 or 1 event can occur (two or more simultaneous events are impossible);

2. the probability of exactly 1 event occurring in $\Delta t$ is equal to $\lambda \Delta t$, where $\lambda$ is a constant; and
3. any non-overlapping intervals of length $\Delta t$ are independent Bernoulli trials,

the probability of $k$ events occurring in the time interval $t = n\Delta t$ is

$$P(k) = \frac{(\lambda t)^k e^{-\lambda t}}{k!}$$

This count distribution is exact for a Poisson process.

In the small-interval Bernoulli approximation, take $p = \lambda \Delta t$. Then

$$P(x) = \frac{(p\frac{t}{\Delta t})^x e^{-p\frac{t}{\Delta t}}}{x!}$$

Next, using the definition $t \equiv n\Delta t$,

$$P(x) = \frac{(pn)^x e^{-pn}}{x!}$$

The interpretation is that for small success probability $p$ and large trial count $n$, $P(x)$ approximates the probability of $x$ successes in $n$ independent trials. The exact finite-trial distribution is binomial.

A common use case for this equation is counting events over observation intervals of length $\Delta t$ (e.g., hurricanes or large solar flares), assuming the Poisson-process conditions hold. The rate parameter $\lambda$ is the expected number of events per unit time, with $p \approx \lambda \Delta t$ for sufficiently short intervals. For example, a rate of 0.01 hurricanes per day means an expected count of 1 hurricane in 100 days, not a guarantee that one will occur.

It also makes sense to talk not about the number of "trials", but rather the number of $\Delta t$s, where each $\Delta t$ corresponds to a trial. In this case, we can define a time as $t = n\Delta t$. This definition allows us to say "given 0.01 hurricanes occur per day, what is the probability that 2 hurricanes occur in a month?".

## 3.2 Continuous

### 3.2.1 Random Variables

A random variable is **continuous** if both of the following apply:

1. Its set of possible values consists either of all numbers in a single interval on the number line (possibly infinite in extent, e.g., from $-\infty$ to $\infty$) or all numbers in a disjoint union of such intervals (e.g., $[0, 10] \cup [20, 30]$).
2. No possible value of the variable has positive probability, that is, $P(X = c) = 0$ for any possible value $c$.

(Devore p 95)

### 3.2.2 PDF Definition

The probability *density* function (pdf) of a continuous random variable $X$ is a function $f(x)$ such that

$$P(a \leq X \leq b) = \int_a^b f(x)dx$$

Important: $f(x)$ is a density so it has units of [units of x]$^{-1}$.

To be a pdf, it must have $f(x) \geq 0$ for all $x$ and $\int_{-\infty}^{\infty} f(x)dx = 1$.

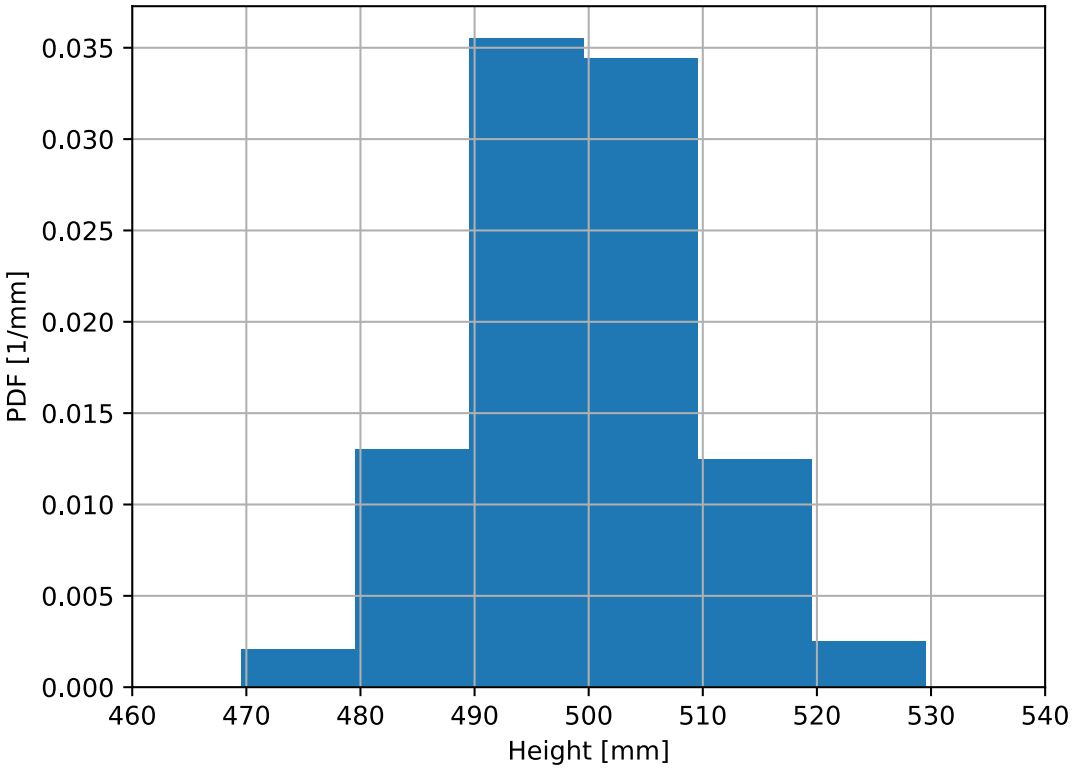
### 3.2.3 Plotting

A PDF need not be continuous. We estimate it from data using a normalized histogram, which is a piecewise constant function.

Code for generating these plots: pdf.py.

---

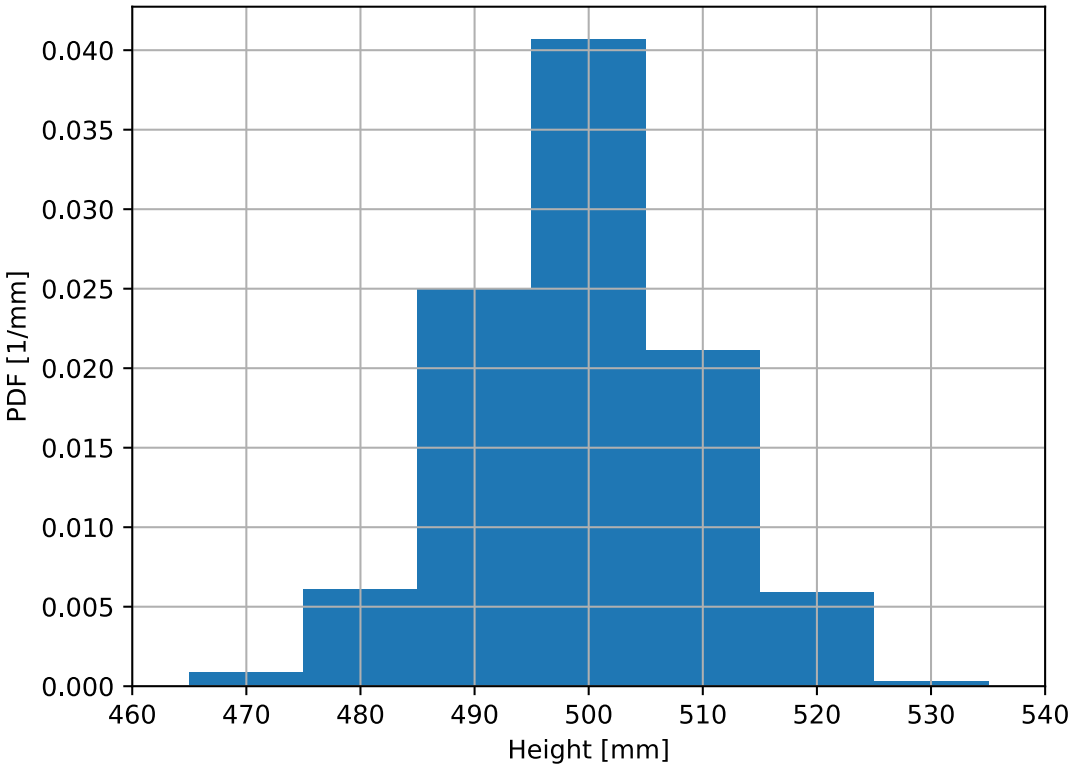
Example of a bad Empirical PDF. Bins are not centered on "nice" numbers.



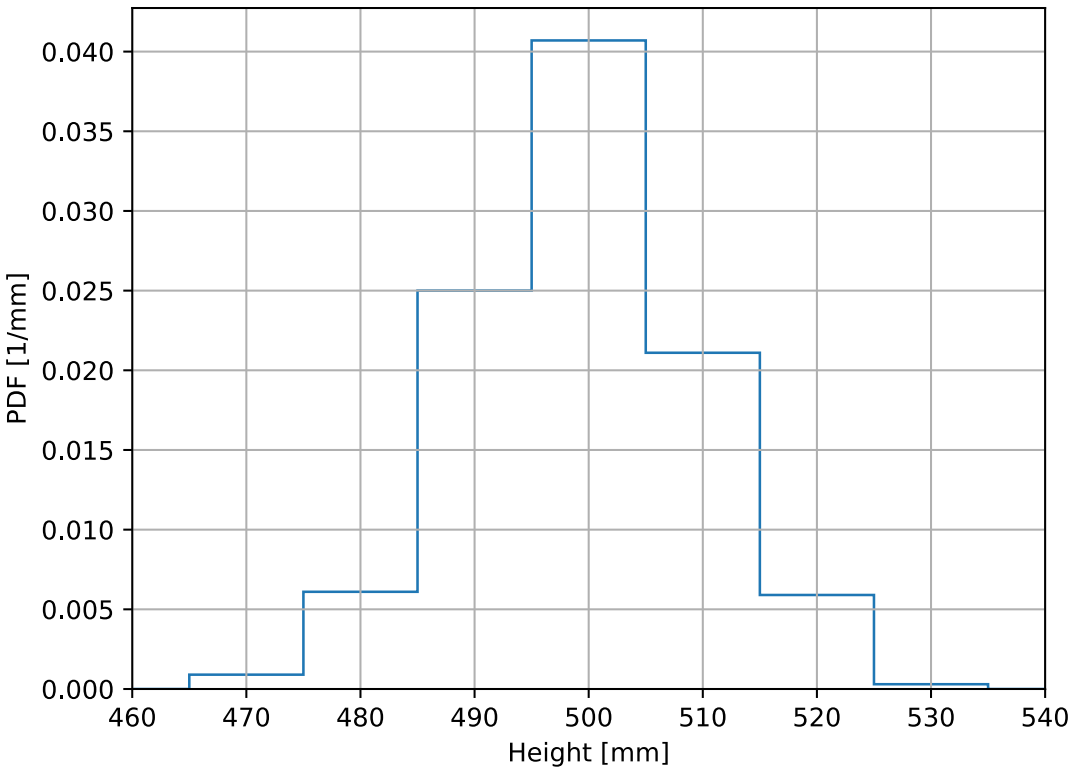

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Example of a good empirical PDF. Bins are centered on "nice" numbers, and bin width is a "nice" number, so I can read off the estimated density for the bin spanning $500 \pm 5$ mm. Note that the vertical axis is not a

probability - if I add the heights of the rectangles, I won't get 1.0 unless the bin width happens to be 1.0.

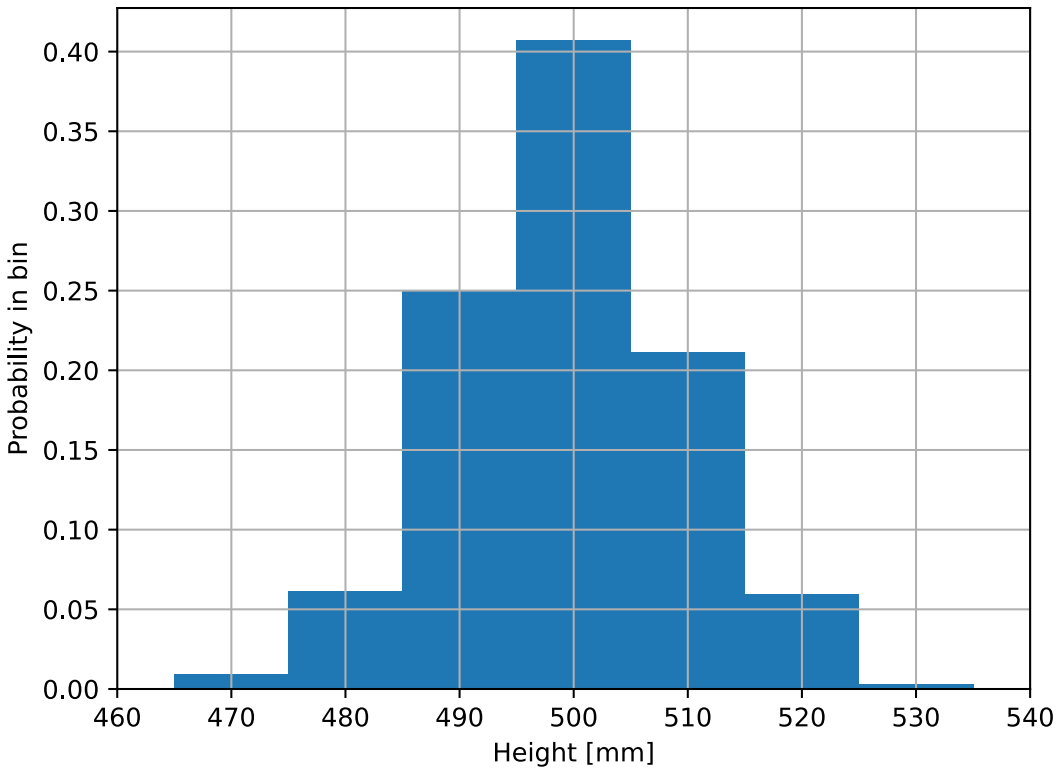


Same as previous, but using `stairs()` plot. This is useful if you are comparing multiple PDFs.



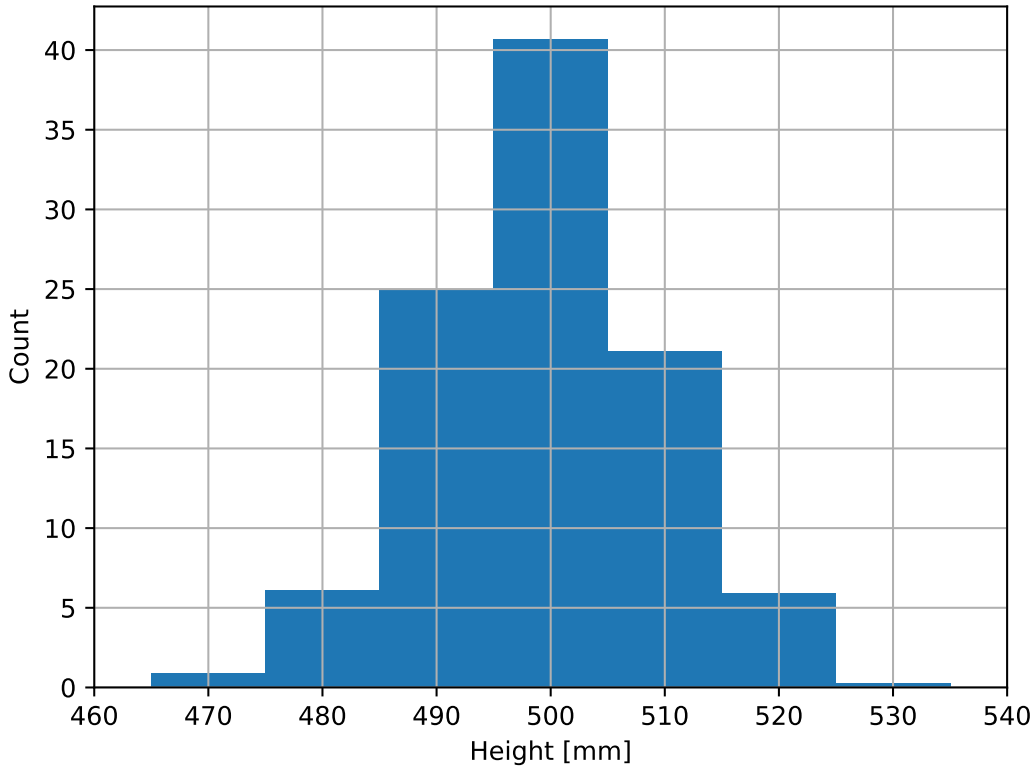

---

Same as previous, but converted to probability in bin by multiplying by bin width.




---

Same as above, but converted from bin probabilities to counts by multiplying by the number of measurements.




### 3.2.4 Expectation Values

$$E[h(X)] = \int_{-\infty}^{\infty} h(x) f(x) dx$$

### 3.2.5 Important PDFs

#### 3.2.5.1 Uniform

$f(x) = 1/(b-a)$ if $a \leq x \leq b$

$f(x) = 0$ otherwise

**Example**

Find $E[X^2]$ for Uniform distribution.

▼ **Answer**

$h(x) = x^2$, so

$$E[X^2] = \int_{-\infty}^{\infty} x^2 f(x) dx$$

$$E[X^2] = \int_{a}^{b} x^2 f(x) dx = \frac{1}{b-a} \int_{a}^{b} x^2 dx = \frac{b^3 - a^3}{3(b-a)}$$

#### 3.2.5.2 Gaussian or Normal

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-(x-\mu)^2/(2\sigma^2)}, \qquad \sigma > 0$$

The “standard normal” is

$$f(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}$$

This is the density of $Z = (X - \mu)/\sigma$.

**Limiting case of Binomial**

As $n \to \infty$, and for $x \approx np$,

$$b(x; n, p) = \binom{n}{x} p^x (1-p)^{n-x} \to \frac{1}{\sqrt{2\pi npq}} e^{-(x-np)^2/2npq}$$

where $q = 1 - p$. Identifying $\sigma^2 = npq$ and $\mu = np$ gives

$$\binom{n}{x} p^x (1-p)^{n-x} \approx \frac{1}{\sigma\sqrt{2\pi}} e^{-(x-\mu)^2/(2\sigma^2)}$$

in the given limits.

▼ **Derivation**

Key steps for $p = 1/2$ to show the approximation:

1. Recast as a random walk problem. Let $n$ be the total number of steps, $x = n_\mathrm{r}$ the number of right steps, and $n_\mathrm{l}$ the number of left steps.
2. Consider the difference $\Delta = n_\mathrm{r} - n_\mathrm{l}$, which corresponds to the distance from the initial position, and assume $\Delta/n \ll 1$. With this definition,
   $$n_\mathrm{r} = \frac{n + \Delta}{2} = \frac{n}{2}\left(1 + \frac{\Delta}{n}\right)$$
   and
   $$n_\mathrm{l} = \frac{n - \Delta}{2} = \frac{n}{2}\left(1 - \frac{\Delta}{n}\right)$$
3. Use Stirling’s approximation $\ln z! \simeq z \ln z - z + \ln\sqrt{2\pi z}$
4. Use $\ln(1+\epsilon) \simeq \epsilon - \dfrac{\epsilon^2}{2}$ for $\epsilon \ll 1$

Recasting the problem using 3.,

$$\ln \binom{n}{x} = \ln \frac{n!}{n_\mathrm{l}!\, n_\mathrm{r}!} = \ln n! - \ln n_\mathrm{l}! - \ln n_\mathrm{r}!$$

Stirling’s approximation on each term is

$$\ln n! = \quad \ln\sqrt{2\pi n} + n\ln n - n$$

$$-\ln n_\mathrm{r}! = -\ln\sqrt{2\pi n_\mathrm{r}} - n_\mathrm{r}\ln n_\mathrm{r} + n_\mathrm{r}$$

$$-\ln n_\mathrm{l}! = -\ln\sqrt{2\pi n_\mathrm{l}} - n_\mathrm{l}\ln n_\mathrm{l} + n_\mathrm{l}$$

The linear terms cancel, since $-n + n_\mathrm{r} + n_\mathrm{l} = 0$. The square-root terms, with $n_\mathrm{r} \simeq n_\mathrm{l} \simeq n/2$, add to

$$\ln\sqrt{2\pi n} - \ln\sqrt{\pi n} - \ln\sqrt{\pi n} = \ln\sqrt{\frac{2}{\pi n}}$$

The inner terms add to

$$n\ln n - n_\mathrm{r}\ln n_\mathrm{r} - n_\mathrm{l}\ln n_\mathrm{l} = (n_\mathrm{r} + n_\mathrm{l})\ln n - n_\mathrm{r}\ln n_\mathrm{r} - n_\mathrm{l}\ln n_\mathrm{l}$$

$$= n_\mathrm{r}\ln n - n_\mathrm{r}\ln n_\mathrm{r} + n_\mathrm{l}\ln n - n_\mathrm{l}\ln n_\mathrm{l}$$

$$= -n_\mathrm{r}\ln\frac{n_\mathrm{r}}{n} - n_\mathrm{l}\ln\frac{n_\mathrm{l}}{n}$$

Using

$$n_\mathrm{r} = \frac{n}{2}\left(1 + \frac{\Delta}{n}\right) \text{ and } n_\mathrm{l} = \frac{n}{2}\left(1 - \frac{\Delta}{n}\right)$$

gives

$$n_\mathrm{r}\ln\frac{n_\mathrm{r}}{n} + n_\mathrm{l}\ln\frac{n_\mathrm{l}}{n} = \frac{n}{2}\left(1 + \frac{\Delta}{n}\right)\ln\left[\frac{1}{2}\left(1 + \frac{\Delta}{n}\right)\right] + \frac{n}{2}\left(1 - \frac{\Delta}{n}\right)\ln\left[\frac{1}{2}\left(1 - \frac{\Delta}{n}\right)\right]$$

The first term

$$\frac{n}{2}\left(1 + \frac{\Delta}{n}\right)\ln\left[\frac{1}{2}\left(1 + \frac{\Delta}{n}\right)\right] = \frac{n}{2}\left(1 + \frac{\Delta}{n}\right)\left[-\ln 2 + \ln\left(1 + \frac{\Delta}{n}\right)\right]$$

The second term

$$\frac{n}{2}\left(1 - \frac{\Delta}{n}\right)\ln\left[\frac{1}{2}\left(1 - \frac{\Delta}{n}\right)\right] = \frac{n}{2}\left(1 - \frac{\Delta}{n}\right)\left[-\ln 2 + \ln\left(1 - \frac{\Delta}{n}\right)\right]$$

Adding, the $\ln 2$ terms gives $-n\ln 2$ and the rest gives

$$n_\mathrm{r}\ln\frac{n_\mathrm{r}}{n} + n_\mathrm{l}\ln\frac{n_\mathrm{l}}{n} = -n\ln 2 + \frac{n}{2}\left[\left(1 + \frac{\Delta}{n}\right)\ln\left(1 + \frac{\Delta}{n}\right) + \left(1 - \frac{\Delta}{n}\right)\ln\left(1 - \frac{\Delta}{n}\right)\right]$$

Using

$$\ln\left(1 + \frac{\Delta}{n}\right) \simeq \frac{\Delta}{n} - \frac{\Delta^2}{2n^2}$$

and

$$\ln\left(1-\frac{\Delta}{n}\right) \simeq -\frac{\Delta}{n} - \frac{\Delta^2}{2n^2}$$

$$n_\mathrm{r}\ln\frac{n_\mathrm{r}}{n} + n_\mathrm{l}\ln\frac{n_\mathrm{l}}{n} \simeq -n\ln 2 + \frac{n}{2}\left[\left(1+\frac{\Delta}{n}\right)\left(\frac{\Delta}{n}-\frac{\Delta^2}{2n^2}\right) + \left(1-\frac{\Delta}{n}\right)\left(-\frac{\Delta}{n}-\frac{\Delta^2}{2n^2}\right)\right]$$

the bracket reduces to $\Delta^2/n^2$ + terms of order $\Delta^3/n^3$ (if a full expansion was performed, we expect only terms of even order in $\Delta$, because the distribution is symmetric about $\Delta$). Thus,

$$n_\mathrm{r}\ln\frac{n_\mathrm{r}}{n} + n_\mathrm{l}\ln\frac{n_\mathrm{l}}{n} \simeq -n\ln 2 + \frac{\Delta^2}{2n}$$

and the inner terms, which carry the opposite sign, give $n\ln 2 - \dfrac{\Delta^2}{2n}$.

So finally

$$\ln\binom{n}{x} \simeq \ln\sqrt{\frac{2}{\pi n}} + n\ln 2 - \frac{\Delta^2}{2n}$$

$$\ln\binom{n}{x} \simeq \ln\left(2^n\sqrt{\frac{2}{\pi n}}e^{-\frac{\Delta^2}{2n}}\right)$$

$$\binom{n}{x} \simeq 2^n\sqrt{\frac{2}{\pi n}}e^{-\frac{\Delta^2}{2n}}$$

Earlier we claimed

$$\binom{n}{x}p^x(1-p)^{n-x} \to \frac{1}{\sqrt{2\pi npq}}e^{-(x-np)^2/2npq}$$

Using $p=q=1/2$, so $npq=n/4$, gives

$$\binom{n}{x}\frac{1}{2^n} \to \frac{1}{\sqrt{2\pi(n/4)}}e^{-(x-n/2)^2/(n/2)}$$

or

$$\binom{n}{x} \to 2^n\sqrt{\frac{2}{\pi n}}e^{-(x-n/2)^2/(n/2)}$$

which agrees with the result above, since $\Delta = 2x-n$ makes $(x-n/2)^2/(n/2) = \Delta^2/2n$.

The derivation for the general case requires the assumption $(x-np)/n$ is small.

#### 3.2.5.3 Student-t

#### 3.2.5.4 $\chi^2$

## 4 Moment Generating Function

(Not covered) A trick that allows one to quickly compute expection values of the form $E[X^k]$. Covered in Bulmer and Larson. The trick is to instead compute $E[e^{tX}]$. Once computed, it is easy to extract $E[X^k]$ from it.

Example:

Let $M(t) = E[e^{tX}]$. Then $dM/dt|_{t=0} = E[X], d^2M/dt^2|_{t=0} = E[X^2], \ldots$.

For the binomial pmf

$$M(t) = E[e^{tX}] = \sum_{x=0}^{n} e^{xt} \binom{n}{x} p^x q^{n-x} = \sum_{x=0}^{n} \binom{n}{x} (pe^t)^x q^{n-x}$$

Using the identity

$$(a+b)^n = \sum_{x=0}^{n} \binom{n}{x} a^x b^{n-x}$$

with $a = pe^t$ and $b = q$ gives

$$M(t) = E\left[e^{tX}\right] = (pe^t + q)^n$$

$$E[X] = \left.\frac{dM}{dt}\right|_{t=0} = n(pe^t + q)^{n-1} pe^t\big|_{t=0} = np(p+q)^{n-1} = np$$

Higher order terms can be computed in a similar way, e.g.,

$$E[X^2] = \left.\frac{d^2M}{dt^2}\right|_{t=0}$$

To understand why this works, consider the Taylor series expansion of $e^{xt}$, which has terms of $xt, (xt)^2, \ldots$.

# 5 Point Estimation

## 5.1 Definition

> A point estimate of a parameter $\theta$ is a single number that can be regarded as a sensible value for $\theta$. A point estimate is obtained by selecting a suitable statistic and computing its value from the given sample data. The selected statistic is called the point estimator of $\theta$. (Devore p 243)

$\hat{\theta}$ is usually a point estimate of a population parameter $\theta$ based on a sample of the population. (Why "point"? Probably because we get a single value.)

Some point estimates have a special name and the hat (^) notation is not used. For example, instead of writing $\hat{\mu}$, we use $\overline{X}$ and instead of $\hat{\sigma^2}$, we write $S^2$.

Point estimators have a sampling distribution – to compute a point estimate, you draw $n$ values from a population (which has a population distribution). If you draw $n$ values again at random, you will not always get the same value for the point estimate.

In the last homework, you numerically estimated the sampling distribution of the point estimates based on samples of size $n$ from $\mathcal{N}(\mu, \sigma^2)$. For the point estimator

- $\overline{X} = (1/n) \sum_{i=1}^{n} x_i$, you found that its sampling distribution had
  - a mean close to $\mu$ (so it seemed unbiased), and
  - a variance that was approximately $\sigma^2/n$ (so its variance decreases as $n$ increases).
- $S_b^2 = (1/n) \sum_{i=1}^{n} (x_i - \overline{X})^2$, you found
  - its mean was not $\sigma^2$. When $n = 10$, the mean was approximately $0.9\sigma^2$. If you re-run the experiment with larger $n$, you will find that the mean of $S_b^2$ will be closer to $\sigma^2$; and
  - its variance was not computed.

In the previous homework you estimated the sampling distribution of the point estimate of $\mu$, $\overline{X}$, by drawing $n$ values from a $\mathcal{N}(\mu, \sigma^2)$ distribution and computing $\overline{X}$. You repeated this $10,000$ times. The mean of the $10,000$ $\overline{X}$s was close to $\mu$ (zero in the homework experiment). The variance of the $10,000$ $\overline{X}$s was approximately $\sigma^2/n$.

In the previous homework you estimated the sampling distribution of the point estimate of $\sigma^2$, $S_b^2$, by drawing $n$ values from a $\mathcal{N}(\mu, \sigma^2)$ distribution and computing $S_b^2$. You repeated this $10,000$ times. The mean of the $10,000$ $S_b^2$s was below $\sigma^2$, near $(n-1)\sigma^2/n$. We did not consider the variance of the $10,000$ $S_b^2$ values.

## 5.2 Some Point Estimators

For the proportion of a population that has a certain characteristic, and $x$ is the number in a sample of $n$ that have that characteristic, we estimate the population proportion, $p$, as

$$\hat{p} = \frac{x}{n}$$

For a sample of $n$ values taken from a population with mean $\mu$, there are several options for $\hat{\mu}$

- $\overline{X} \equiv \dfrac{1}{n} \displaystyle\sum_{i=1}^{n} x_i$
- $\widetilde{X} \equiv$ median (middle value when $n$ is odd or average of two middle values when $n$ is even)
- $X_e \equiv (\min(x) + \max(x))/2$ (average of extreme values from sample)
- $\overline{X}_{\text{tr}(m)} \equiv$ trimmed mean, the mean after removing the largest $m\%$ and smallest $m\%$ values from the sample.

For a sample of $n$ values taken from a population with variance $\sigma^2$, there are several "sensible" options for $\hat{\sigma^2}$

$$S_b^2 \equiv \frac{1}{n} \sum_{i=1}^{n} (x_i - \overline{X})^2$$

$$S^2 \equiv \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \overline{X})^2$$

## 5.3 Choosing a Point Estimate

A given population parameter may have more than one "sensible" point estimator.

We want point estimators (which are random variables, so they have a distribution) to have the properties:

1. Unbiased – The expected value of the sampling distribution of the point estimate is the population parameter.
2. Small variance – The variance of the sampling distribution of the point estimate is small.

Formal definition of bias:

> A point estimator $\hat{\theta}$ is said to be an unbiased estimator of $\theta$ if $E[\hat{\theta}] = \theta$ for every possible value of $\theta$. If it is not unbiased, the difference is called the bias of $\hat{\theta}$. (Devore p 243)

**Examples**:

- If $n$ independent Bernoulli trials have success probability $p$ and $x$ is the number of successes, then $\hat{p} = x/n$ is an unbiased estimator of $p$ (Devore p 244, but stated in a different way).
- If $n$ values are drawn from *any* distribution (continuous or discrete) with a finite mean $\mu$, $\overline{X}$ is an unbiased estimator of $\mu$ (based on Devore p 246).
- For an independent random sample from a continuous distribution symmetric about $\mu$, the sample median and symmetrically trimmed means are unbiased estimators of $\mu$, provided their expectations exist (based on Devore p 246).

Given that multiple sensible point estimates exist, which should we choose? It depends on the population distribution. In general, we prefer unbiased estimators with small variance. However, sometimes an estimator is biased but has a small variance, and we may prefer it over one that has no bias but a large variance. See Example 6.7 of Devore.

For some population models and parameters, we know an estimator that is unbiased and has the smallest variance among *all unbiased estimators* of that parameter. This is called the Minimum Variance Unbiased Estimator, **MVUE**. For example, for an independent random sample from a Gaussian (normal) population, $\overline{X}$ is the MVUE of $\mu$. Identifying an MVUE depends on the population model and the parameter being estimated.

## 5.4 Point Estimate Bias and Variance

To compute an estimate, such as an observed value of $\overline{X}$, we draw random values from a population. Each draw is modeled by a random variable; the population itself is not a random variable.

The estimator itself is a random variable because when we randomly choose a set of $n$ values and compute the estimator, we won't get the same result when we randomly choose another set of $n$ values. You demonstrated this on the last homework by plotting a histogram of 10,000 $\overline{X}$ values. This histogram represented the **sampling distribution** of $\overline{X}$.

We don't always know the sampling distribution of an estimator. We only know the analytical value for a few cases. For example, if the population is Gaussian distributed ($X \sim \mathcal{N}(\mu, \sigma^2)$), then the sampling distribution is normally distributed with a smaller variance ($\overline{X} \sim \mathcal{N}(\mu, \sigma^2/n)$). You demonstrated this numerically on the last homework.

Proving that an estimator is unbiased does not establish that it has minimum variance. Identifying an MVUE requires a separate argument.

## 5.5 Numerically Estimating Sampling Distribution

Numerically generate the sampling distribution of $\overline{X}_{\text{tr}(10)}$ and compare it to $\overline{X}$.

## 5.6 Computing Bias

### 5.6.1 $\overline{X}$

We can show that $\overline{X}$ is an unbiased estimator of $\mu$, that is, $E[\overline{X}] = \mu$, for any population distribution with a finite mean.

$$\overline{X} = \frac{1}{n}\sum_{i=1}^{n} X_i$$

Taking the expectation, we have

$$E[\overline{X}] = E\left[\frac{1}{n}\sum_{i=1}^{n} X_i\right]$$

For a discrete random variable, note that

$$E\left[X\right] = \sum_{\text{all } x} xP(x)$$

The corresponding expectation for a continuous random variable is an integral. In either case, linearity of expectation gives

$$E[\overline{X}] = \frac{1}{n}\sum_{i=1}^{n} E[X_i] = \frac{1}{n}(E[X_1] + E[X_2] + ...)$$

The term $E[X_1]$ is the expected value of the first draw. Each draw has the same distribution as $X$, so $E[X_i] = E[X] = \mu$ for every $i$. Thus, we have

$$E[\overline{X}] = \frac{1}{n}\sum_{i=1}^{n} E[X_i] = \frac{1}{n}(E[X_1] + E[X_2] + ...) = \frac{1}{n}(\mu + \mu + ...) = \frac{1}{n}(n\mu) = \mu$$

### 5.6.2 $S_b^2$

For $n$ independent, identically distributed draws from a population with finite variance $\sigma^2$, we can show that $E\left[S_b^2\right] = \sigma^2(n-1)/n$.

In the following, I give a very verbose proof to help you understand the operations and concepts involved. Devore p. 245 has a briefer proof that is given below.

#### 5.6.2.1 Long proof

Starting with the definition of $S_b^2$,

$$E\left[S_b^2\right] = E\left[\frac{1}{n}\sum_{i=1}^{n}(X_i - \overline{X})^2\right]$$

Expanding the square gives

$$E\left[S_b^2\right] = E\left[\frac{1}{n}\sum_{i=1}^{n}(X_i^2 - 2X_i\overline{X} + \overline{X}^2)\right]$$

or, equivalently,

$$I. \qquad E\left[S_b^2\right] = E\left[\frac{1}{n}\sum_{i=1}^{n}X_i^2 - \frac{2}{n}\sum_{i=1}^{n}X_i\overline{X} + \frac{1}{n}\sum_{i=1}^{n}\overline{X}^2\right]$$

The second term in the square braces in Equation $I.$ can be simplified by noting that $\overline{X}$ does not depend on $i$, so it can be factored out:

$$\frac{2}{n}\sum_{i=1}^{n}X_i\overline{X} = 2\overline{X}\left(\frac{1}{n}\sum_{i=1}^{n}X_i\right)$$

Using this and the definition $\overline{X} \equiv (1/n)\sum_{i=1}^{n}X_i$, the second term in Equation $I.$ is

$$\frac{2}{n}\sum_{i=1}^{n}X_i\overline{X} = 2\overline{X}^2$$

The third term is

$$\frac{1}{n}\sum_{i=1}^{n}\overline{X}^2 = \frac{1}{n}n\overline{X}^2 = \overline{X}^2$$

Therefore, the second and third terms combine to be $-2\overline{X}^2 + \overline{X}^2 = -\overline{X}^2$ and Equation $I.$ simplifies to

$$E\left[S_b^2\right] = E\left[\frac{1}{n}\sum_{i=1}^{n}X_i^2 - \overline{X}^2\right]$$

Distributing the $E$ to each of the two terms by using $E[a+b] = E[a] + E[b]$ gives

$$E\left[S_b^2\right] = E\left[\frac{1}{n}\sum_{i=1}^{n}X_i^2\right] - E\left[\overline{X}^2\right]$$

or

$$E\left[S_b^2\right] = \frac{1}{n}\left(E[X_1^2] + E[X_2^2] + ...\right) - E\left[\overline{X}^2\right]$$

Note that $E[X_1^2]$ means "the expectation of the square of the first value of the sample". Each draw has the same distribution as $X$, so $E[X_1^2] = E[X^2]$. Using this, we have

$II.$ $\quad E\left[S_b^2\right] = E\left[X^2\right] - E\left[\overline{X}^2\right]$

Both of the terms in Equation $II.$ can be re-written using $\mu$, $\sigma$, and $n$.

The first term simplifies to

$E\left[X^2\right] = \mu^2 + \sigma^2,$

which follows from the definition

$\sigma^2 \equiv E\left[(X-\mu)^2\right]$, expanding the square and using $E[a+b] = E[a] + E[b]$ to give

$\sigma^2 = E\left[X^2 - 2\mu X + \mu^2\right] = E\left[X^2\right] - 2\mu E[X] + E[\mu^2]$

Using the definition $\mu \equiv E[X]$ and the fact that $E[\mu^2] = \mu^2$ because $\mu$ is a constant, this simplifies to

$\sigma^2 = E[X^2] - \mu^2$

giving $E\left[X^2\right] = \mu^2 + \sigma^2$ as claimed.

Therefore, the first term in Equation $II.$ is

$$\frac{1}{n}\sum_{i=1}^{n} E\left[X_i^2\right] = \frac{1}{n}\sum_{i=1}^{n}(\mu^2+\sigma^2) = \frac{1}{n}n(\mu^2+\sigma^2) = \mu^2+\sigma^2$$

The second term in Equation $II.$, $E[\overline{X}^2]$, can be rewritten as

$$E\left[\overline{X}^2\right] = E\left[\left(\frac{1}{n}\sum_{i=1}^{n} X_i\right)^2\right] = \frac{1}{n^2}E\left[\left(\sum_{i=1}^{n} X_i\right)^2\right]$$

The square of the sum,

$$\left(\sum_{i=1}^{n} X_i\right)^2 = (X_1 + X_2 + ...)(X_1 + X_2 + ...)$$

expands to

$X_1^2 + X_1(X_2 + X_3 + ... + X_n) +$

$X_2^2 + X_2(X_1 + X_3 + X_4 + ... + X_n) +$

...

$X_n^2 + X_n(X_1 + X_2 + ... + X_{n-1})$

Let $X' = X - \mu$. Then, the above can be written as

$(X_1' + \mu)^2 + (X_1' + \mu)(X_2' + \mu + X_3' + \mu + ... + X_n' + \mu) +$

$(X_2' + \mu)^2 + (X_2' + \mu)(X_1' + \mu + X_3' + \mu + ... + X_n' + \mu) +$

...

$(X'_n + \mu)^2 + (X'_n + \mu)(X'_1 + \mu + X'_2 + \mu + ... + X'_{n-1} + \mu)$

Next we note that $E\left[X'_i X'_j\right] = 0$ for $i \neq j$; this is due to the samples being independent, so $E[X'_i X'_j] = E[X'_i]E[X'_j]$ and the fact that $E[X'] = E[X] - E[\mu] = 0$

**Problems**

1. Show that
   $E\left[X'\right] = 0$,
   $E\left[X'^2\right] = \sigma^2$.
2. Finish the proof that $E\left[S_b^2\right] = \sigma^2(n-1)/n$.

**Answer**

1. $E[X'] = E[X - \mu] = E[X] - \mu = \mu - \mu = 0$
   $E\left[X'^2\right] = E\left[X^2 - 2\mu X + \mu^2\right] = E[X^2] - E[2\mu X] + E[\mu^2] = E[X^2] - 2\mu^2 + \mu^2 = E[X^2] - \mu^2$
   Earlier it was shown that $E\left[X^2\right] = \mu^2 + \sigma^2$, so substitution gives
   $E\left[X'^2\right] = \sigma^2$
2. Consider the following rows of the expansion of $\left(\sum_{i=1}^n X_i\right)^2$, starting with the first row:
   $(X'_1 + \mu)^2 + (X'_1 + \mu)(X'_2 + \mu + X'_3 + \mu + ... + X'_n + \mu) +$
   $(X'_2 + \mu)^2 + (X'_2 + \mu)(X'_1 + \mu + X'_3 + \mu + ... + X'_n + \mu) +$
   ...
   $(X'_n + \mu)^2 + (X'_n + \mu)(X'_1 + \mu + X'_2 + \mu + ... + X'_{n-1} + \mu)$
   Its first term, $(X'_1 + \mu)^2$, is $X_1^2$, which has an expectation value of $E[X^2]$, which was shown earlier to be $\mu^2 + \sigma^2$.
   In the second term,
   $(X'_1 + \mu)(X'_2 + \mu + X'_3 + \mu + ... + X'_n + \mu)$,
   there are $n - 1$ $\mu$ terms, so expanding gives
   $\sum_{j=2}^n X'_1 X'_j + (n-1)\mu X'_1 + \mu \sum_{j=2}^n X'_j + (n-1)\mu^2$
   Because $E[X'_i X'_j] = 0$ for $i \neq j$ and $E[X'_i] = 0$,
   $E\left[\sum_{j=2}^n X'_1 X'_j + (n-1)\mu X'_1 + \mu \sum_{j=2}^n X'_j + (n-1)\mu^2\right] = (n-1)\mu^2$
   Thus,
   $E\left[(X'_1 + \mu)^2 + (X'_1 + \mu)(X'_2 + \mu + X'_3 + \mu + ... + X'_n + \mu)\right] = \mu^2 + \sigma^2 + \mu^2(n-1) = \sigma^2 + n\mu^2$
   We only considered one row. There are a total of $n$ rows, so

$$E\left[\overline{X}^2\right] = \frac{1}{n^2} E\left[\left(\sum_{i=1}^{n} X_i\right)^2\right] = \frac{1}{n^2}(n\sigma^2 + n^2\mu^2) = \mu^2 + \sigma^2/n$$

In summary, we have shown that $E[X^2] = \mu^2 + \sigma^2$ and $E[\overline{X}^2] = \mu^2 + \sigma^2/n$. As a result, Equation $II$. simplifies to

$$E[S_b^2] = E\left[X^2\right] - E\left[\overline{X}^2\right] = \mu^2 + \sigma^2 - \left(\mu^2 + \sigma^2/n\right) = \sigma^2 - \sigma^2/n$$

or,

$$E[S_b^2] = \frac{n-1}{n}\sigma^2$$

with the interpretation that $S_b^2$ is downward biased by a factor of $(n-1)/n$.

This equation makes sense in one limit - suppose we draw a single sample from a distribution with variance $\sigma^2$. $S_b^2$ will always be zero, which is less than $\sigma^2$ for any nonzero $\sigma$.

It follows from the calculation above that an unbiased estimate of the variance of a population is

$$S^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \overline{X})^2$$

because $E[S^2] = \sigma^2$. Note that when $n = 1$, $S^2$ is $0/0$, which is indeterminate; this makes sense, as we don't expect to be able to estimate the variance of a population with only one sample.

Recall that, in contrast, an unbiased estimate of $\mu$ is

$$\overline{X} = \frac{1}{n}\sum_{i=1}^{n} X_i$$

which has $n$ and not $n-1$.

#### 5.6.2.2 Alternative proof 1. (Devore)

Devore on p 245 starts with the observation that the variance of a random variable $Y$, $V(Y)$, can be written

$$V(Y) = E[Y^2] - (E[Y])^2$$

or, rearranging,

$$A. \qquad E[Y^2] = V(Y) + (E[Y])^2,$$

which is stated without proof. The formula is also given on page 112, but a proof is not given.

Next, he states that it follows from the definition

$$S^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \overline{X})^2$$

that

$$S^2 = \frac{1}{n-1}\left[\sum_{i=1}^{n} X_i^2 - \frac{1}{n}\left(\sum_{i=1}^{n} X_i\right)^2\right].$$

(This follows from $\sum(X_i - \overline{X})^2 = \sum(X_i^2 - 2X_i\overline{X} + \overline{X}^2) = \sum X_i^2 - 2n\overline{X}^2 + n\overline{X}^2 = \sum X_i^2 - n\overline{X}^2$ and $\overline{X} = (1/n)\sum X_i$.)

Taking the expectation and moving it inside of the sum in the first term gives

$$E[S^2] = \frac{1}{n-1}\left(\sum_{i=1}^{n} E[X_i^2] - \frac{1}{n}E\left[\left(\sum_{i=1}^{n} X_i\right)^2\right]\right)$$

Using equation $A.$, which is $E[Y^2] = V(Y) + (E[Y])^2$, with $Y = X_i$ is
$E[X_i^2] = V(X_i) + (E[X_i])^2 = \sigma^2 + \mu^2$ using the definitions of $\sigma$ and $\mu$. Using this, the above equation can be re-written as

$$E[S^2] = \frac{1}{n-1}\left(\sum_{i=1}^{n} (\sigma^2 + \mu^2) - \frac{1}{n}E\left[\left(\sum_{i=1}^{n} X_i\right)^2\right]\right)$$

Using equation $A.$ again, which is $E[Y^2] = V(Y) + (E[Y])^2$, this time with $Y = \sum_{i=1}^{n} X_i$ gives

$$E\left[\left(\sum_{i=1}^{n} X_i\right)^2\right] = V\left(\sum_{i=1}^{n} X_i\right) + \left(E\left[\sum_{i=1}^{n} X_i\right]\right)^2 = n\sigma^2 + (n\mu)^2$$

so now

$$E[S^2] = \frac{1}{n-1}\left(\sum_{i=1}^{n} (\sigma^2 + \mu^2) - \frac{1}{n}\left[n\sigma^2 + (n\mu)^2\right]\right)$$

or

$$E[S^2] = \frac{1}{n-1}\left(n(\sigma^2 + \mu^2) - \frac{1}{n}(n\sigma^2 + n^2\mu^2)\right) = \sigma^2$$

### 5.6.3 $S^2$

For an independent random sample with $n > 1$ from any population with finite variance, we can show that $S^2$ is an unbiased estimator of $\sigma^2$ using the result from $S_b^2$.

### 5.6.4 $S$

Computing an unbiased estimator of $\sigma$ from $S = \sqrt{S^2}$ is more difficult than estimating $\sigma^2$ with $S^2$. With the restriction that the population is normally distributed with standard deviation $\sigma$, we have a formula discussed in Problem 37 on page 266 of Devore (I don't have a reference for the proof):

> 37. When the sample standard deviation $S$ is based on a random sample from a normal population distribution, it can be shown that

$$E[S] = \sigma\sqrt{\frac{2}{n-1}}\frac{\Gamma\left(\frac{n}{2}\right)}{\Gamma\left(\frac{n-1}{2}\right)}$$

Use this to obtain an unbiased estimator for $\sigma$ of the form $cS$. What is $c$ when $n = 20$?

For positive integer $x$, the gamma function $\Gamma(x) = (x-1)!$, so $\Gamma(10) = 9 \cdot 8 \cdot ... \cdot 1 = 362880$.

For positive non-integer $x$, the gamma function can be evaluated numerically; half-integer values can also be computed using $\Gamma(1/2) = \sqrt{\pi}$ and $\Gamma(x+1) = x\Gamma(x)$. Using a gamma function calculator gives $\Gamma(9.5) \approx 119292.5$. Alternatively, for large $x$, $\Gamma(x+1) \sim \sqrt{2\pi x}\left(\frac{x}{e}\right)^x$. Using this with $x = 8.5$ gives $\Gamma(9.5) \approx 118129.2$.

To be unbiased, we want

$cE[S] - \sigma = 0$. Solving for $c$ gives

$$c = \sqrt{\frac{n-1}{2}}\frac{\Gamma\left(\frac{n-1}{2}\right)}{\Gamma\left(\frac{n}{2}\right)}$$

Plugging in values,

$$c \approx \sqrt{9.5}\frac{119292.5}{362880} \approx 1.01$$

### 5.6.5 Linear regression slope parameter

In your physics labs, you use the equation

$$b = \frac{\sum_{i=1}^{n} x_i y_i - n\bar{x}\bar{y}}{\sum_{i=1}^{n} x_i^2 - n\bar{x}^2} = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^{n}(x_i - \bar{x})^2}$$

as an estimate of the population parameter $\beta$ in the model equation

$y_i = \beta x_i + \alpha + \epsilon_i$

where $\epsilon_i$ are independent and randomly distributed values from a Gaussian distribution with zero mean and standard deviation $\sigma$.

Assume the $x_i$ values are fixed and not all equal, so $\sum_{i=1}^{n}(x_i - \bar{x})^2 > 0$. Then $x_i - \bar{x}$ can be treated as a constant. Thus

$$E[b] = E\left[\frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^{n}(x_i - \bar{x})^2}\right] = \frac{E\left[\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})\right]}{\sum_{i=1}^{n}(x_i - \bar{x})^2}$$

and

$$E[b] = \frac{\sum_{i=1}^{n}(x_i - \bar{x})E\left[(y_i - \bar{y})\right]}{\sum_{i=1}^{n}(x_i - \bar{x})^2}$$

$$E\left[y_i - \bar{y}\right] = E\left[\beta x_i + \alpha + \epsilon_i - \bar{y}\right] = E[\beta x_i] + E[\alpha] + E[\epsilon_i] - E[\bar{y}]$$

$$E\left[y_i - \bar{y}\right] = \beta x_i + \alpha - E[\overline{y}]$$

$$E[\overline{y}] = E[\beta\overline{x} + \alpha + \overline{\epsilon}] = \beta\overline{x} + \alpha + E[\overline{\epsilon}] = \beta\overline{x} + \alpha$$

Thus,

$$E\left[y_i - \bar{y}\right] = \beta(x_i - \overline{x})$$

and so

$$E[b] = \beta$$

Alternative:

Substitution of

$$y_i = \beta x_i + \alpha + \epsilon_i$$

into

$$b = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^{n}(x_i - \bar{x})^2}$$

gives

$$b = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(\beta x_i + \alpha + \epsilon_i - \bar{y})}{\sum_{i=1}^{n}(x_i - \bar{x})^2}$$

Using

$$\overline{y} = \alpha + \beta\overline{x} + \overline{\epsilon}$$

gives

$$b = \frac{\sum_{i=1}^{n}(x_i - \bar{x})\left[\beta(x_i - \overline{x}) + \epsilon_i - \bar{\epsilon}\right]}{\sum_{i=1}^{n}(x_i - \bar{x})^2}$$

Since $\sum_{i=1}^{n}(x_i - \bar{x}) = 0$, the term involving $\bar{\epsilon}$ cancels, giving

$$b = \frac{\sum_{i=1}^{n}\left[\beta(x_i - \overline{x})^2 + (x_i - \overline{x})\epsilon_i\right]}{\sum_{i=1}^{n}(x_i - \bar{x})^2} = \beta + \frac{\sum_{i=1}^{n}(x_i - \overline{x})\epsilon_i}{\sum_{i=1}^{n}(x_i - \bar{x})^2}$$

Finally, using

$$E\left[\frac{\sum_{i=1}^{n}(x_i - \overline{x})\epsilon_i}{\sum_{i=1}^{n}(x_i - \bar{x})^2}\right] = \frac{\sum_{i=1}^{n}(x_i - \overline{x})E[\epsilon_i]}{\sum_{i=1}^{n}(x_i - \bar{x})^2} = 0$$

because the $x$ values are fixed and can be treated as constants, we can conclude that $E[b] = \beta$.

# 6 Point Estimate Intervals

- Confidence Intervals
- Confidence Bounds (not covered; p. 282 of Devore)
- Prediction Intervals (not covered; p. 289 of Devore)
- Tolerance Intervals (not covered; p. 291 of Devore)

# 7 Confidence Intervals

> An alternative to reporting a single sensible value for the parameter being estimated is to calculate and report an entire interval of plausible values—an interval estimate or confidence interval (CI). A confidence level of 95% implies that 95% of all samples would give an interval that includes $\mu$, or whatever other parameter is being estimated, and only 5% of all samples would yield an erroneous interval. The most frequently used confidence levels are 95%, 99%, and 90%. The higher the confidence level, the more strongly we believe that the value of the parameter being estimated lies within the interval (an interpretation of any particular confidence level will be given shortly). (Devore p 267)

## 7.1 Interpretation

If instead of taking one sample and computing a CI, you took many samples and computed many CIs for a sample statistic (repeated the experiment), the percentage of the CIs that would "trap" the value of the population that the sample statistic approximates is the confidence level.

The following figure is from Devore 8th Edition. In this case the sample statistic is $\overline{x}$, which is an estimate of the population mean $\mu$. Each horizontal line is computed from the $\overline{x}$ of a sample plus its CI.

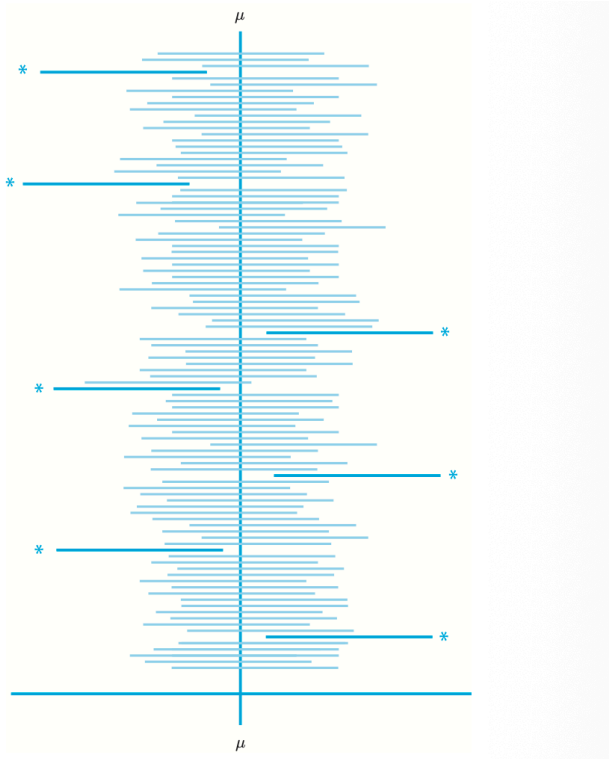



**Figure 7.3** One hundred 95% CIs (asterisks identify intervals that do not include $\mu$).

## 7.2 Misinterpretation

A common misinterpretation of a 95% CI is "there is a 95% chance that the population value (such as $\mu$) is in the CI of the sample statistic that is used to estimate the population value (such as $\overline{x}$)".

To see the problem with this, consider what would happen if experimenter 1 and experimenter 2 obtain confidence intervals of

1. $[-1.0, 0.0]$
2. $[0.0, +1.0]$

If both of them claim that there is a 95% chance $\mu$ is in their CI, there is a contradiction.

Another justification for why the quote is wrong is when computing a CI, we assumed the true population parameter is a fixed, unknown constant, and a calculated CI either contains it (probability = 1) or does not contain it (probability = 0).

Another misinterpretation is that "overlapping CIs mean no significant difference". (I did not cover this; determining if this conclusion is appropriate or not requires the hypothesis test methods covered in Chapter 9 of Devore.)

Note that there is an alternative to a confidence interval called a credible interval. With a credible interval, one can make a claim about the probability that the population value is in the range of a credible interval. This is a topic in Bayesian inference that is covered later. In the Bayesian method, the population parameter is not regarded as a fixed unknown value but rather as a unknown value that has a probability distribution.

## 7.3 Deriving

Deriving a confidence interval requires deriving the sampling distribution of a point estimator. This is not covered in these notes.

## 7.4 Computing

Requires

1. Assumption about distribution of population
2. Finding the appropriate CI formula for the point estimator

Formulas depend on

1. Population distribution (e.g., normal or Student's $t$),
2. Known parameters of the population distribution, if any (e.g., $\sigma$ known),
3. Number of samples, $n$
4. The confidence level $1-\alpha$ (e.g., 0.95 when $\alpha = 0.05$)

## 7.5 Common CIs

### 7.5.1 Normal Mean, Known Standard Deviation

A $100(1-\alpha)\%$ confidence interval for the mean $\mu$ of a normal population when the value of $\sigma$ is known is given by

$$\left(\overline{x} - z_{\alpha/2}\frac{\sigma}{\sqrt{n}}, \quad \overline{x} + z_{\alpha/2}\frac{\sigma}{\sqrt{n}}\right)$$

Devore Equation 7.5

Devore does not define $\alpha$ until chapter 8 where it is defined as a signficance level. This may be confusing because the equation implies a large confidence interval corresponds to a low significance level. However, we have not defined what significance is with respect to. Based on the definition of the confidence interval, repeated experiments will result in the population mean being outside of the confidence interval in a fraction, $\alpha$, of experiments. So the interpretation of "low significance" means that the probability of an experiment yielding a confidence interval that does not cover the true mean is "insignificant" in the sense that the probability is a small number $\alpha$.

**Example**

```
import numpy as np
np.random.seed(0)
n = 10
μ = 0
σ = 2
x = np.random.normal(loc=μ, scale=σ, size=n)

print(x)
# [ 3.52810469  0.80031442  1.95747597  4.4817864   3.73511598
   -1.95455576  1.90017684 -0.30271442 -0.2064377   0.821197  ]

print(np.mean(x))
# 1.4760463414576694
```

Compute a 95% CI for $\mu$ using this observed $\overline{X}$.

This means $\alpha = 0.05$. First, find $z_{\alpha/2} = z_{0.025}$. The interpretation of $z_A$ is that an area of $A$ under the standard normal PDF is to the left of $-z_A$.

Standard Normal PDF:

$$f(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}$$

### 7.5.2 Normal Mean, Unknown Standard Deviation

Let $\overline{x}$ and $s$ be the sample mean and the sample standard deviation computed from the results of a random sample from a normal population with a mean $\mu$. Then a $100(1-\alpha)\%$ confidence interval for the mean $\mu$ is

$$\left( \overline{x} - t_{\alpha/2,n-1} \frac{s}{\sqrt{n}}, \quad \overline{x} + t_{\alpha/2,n-1} \frac{s}{\sqrt{n}} \right)$$

Devore Equation 7.15, p 288

### 7.5.3 Approximately Normal Estimator

Suppose $\hat{\theta}$ is an estimator satisfying the following properties: (1) It has approximately a normal distribution; (2) it is (at least approximately) unbiased; and (3) an expression for $\sigma_{\hat{\theta}}$, the standard deviation of $\hat{\theta}$, is available. Then

$$P\left(-z_{\alpha/2} < \frac{\hat{\theta} - \theta}{\sigma_{\hat{\theta}}} < z_{\alpha/2}\right) \simeq 1 - \alpha$$

Devore Equation 7.9

### 7.5.4 Population Proportion

Devore Equation 7.10 gives a complex formula for the CI for a population proportion, $p$. For large $n$, it is

$$\hat{p} \pm z_{\alpha/2}\sqrt{\hat{p}\hat{q}/n}$$

Devore page 280.

For small $n$, there are many complications. See discussion in Devore p 281 and Brown et al., Interval Estimation for a Binomial Proportion, 2001.

### 7.5.5

A $100(1-\alpha)\%$ confidence interval for the variance $\sigma^2$ of a normal population has a lower limit

$$S^2(n-1)/\chi^2_{\alpha/2,n-1}$$

and an upper limit

$$S^2(n-1)/\chi^2_{1-\alpha/2,n-1}$$

A confidence interval for $\sigma$ has lower and upper limits that are the square roots of the corresponding limits in the interval for $\sigma^2$.

Devore page 295 (box on bottom of page)

Recall that $\alpha$ is referred to as the significance level.

Note that $\chi^2_{\alpha/2,n-1}$ corresponds to the value of $\chi^2_{n-1}$ such that the area *to the right* is $\alpha/2$. This is opposite of $|z_{\alpha/2}|$, which is $|z|$ such that the area *to the left* is $\alpha/2$.

It is best to always think of the $\alpha/2$ values as corresponding to a small area.

# 8 Hypothesis Tests

## 8.1 Overview

The null hypothesis, denoted by $H_0$, is the claim that is initially assumed to be true (the "prior belief" claim). The alternative hypothesis, denoted by $H_a$, is the assertion that is contradictory to $H_0$.

The null hypothesis will be rejected in favor of the alternative hypothesis only if sample evidence suggests that $H_0$ is false. If the sample does not strongly contradict $H_0$, we will continue to believe in the plausibility of the null hypothesis. The two possible conclusions from a hypothesis-testing analysis are then reject $H_0$ or fail to reject $H_0$. (Devore p. 301)

$H_0$ will generally be stated as an equality claim. If $\theta$ denotes the parameter of interest, the null hypothesis will have the form $H_0$: $\theta = \theta_0$, where $\theta_0$ is a specified number called the null value of the parameter (value claimed for $\theta$ by the null hypothesis). (Devore p. 302)

A test procedure is specified by the following:

1. A test statistic, a function of the sample data on which the decision (reject $H_0$ or do not reject $H_0$) is to be based [example $\overline{x}$]
2. A rejection region, the set of all test statistic values for which $H_0$ will be rejected

The null hypothesis will then be rejected if and only if the observed or computed test statistic value falls in the rejection region. (Devore p. 303)

A type I error consists of rejecting the null hypothesis $H_0$ when it is true.

A type II error involves not rejecting $H_0$ when $H_0$ is false.

The choice of a particular region cutoff values fixes the probabilities of type I and type II errors. These error probabilities are traditionally denoted by $\alpha$ and $\beta$, respectively.

(Devore p. 304)

One can also define rejection regions based on an assumed value of $\alpha$. That is, given $\alpha$, one can compute the rejection region.

## 8.2 Basic Rejection Region Algorithm

The following applies to a problem where one wants to test for equality with an alternative of inequality.

1. Choose a population parameter, $\theta$ that you want to subject to an equality claim ("null hypothesis", $H_0$) of the form $\theta = \theta_0$, where $\theta_0$ is the "null value".

   For example, suppose we want to test the claim that the mean of a population is 1.0; then $\theta = \mu$ and $\theta_0 = 1.0$ and $H_0$ is $\mu = 1.0$.

2. Find the equation for $\hat{\theta}$, an unbiased estimator of $\theta$.

   For example, if we want to test the claim that $\mu = 1.0$, an unbiased estimator of $\mu$ is $\overline{x}$, so $\hat{\theta} = \overline{x}$.

3. Determine the sampling distribution of $\hat{\theta}$. This sampling distribution will depend on the population distribution (which is rarely known). In most textbook examples, the exact sampling distribution is known because the population distribution is assumed to be known. Also, textbook sampling distributions for $\hat{\theta}$ are usually from a short list of common distributions: $\mathcal{N}, t, \chi^2, F$.

   Examples:

   - If population is $\mathcal{N}(\mu, \sigma)$, the sampling distribution of $\hat{\theta} = \overline{x}$ is $\mathcal{N}(\mu, \sigma/\sqrt{n})$. Or, equivalently, the sampling distribution of $(\overline{x} - \mu)/(\sigma/\sqrt{n})$ is $\mathcal{N}(\mu = 0, \sigma = 1)$.
   - If population is $\mathcal{N}(\mu, \sigma)$, the sampling distribution of $\hat{\theta} = s^2$ can be determined using the fact that $s^2(n-1)/\sigma^2$ is $\chi^2_{n-1}$. (In contrast to the previous case, there is not a common notation that is used to say, for example, "$s^2$ has a sampling distribution of $\chi^2_{n-1}(\sigma^2)$".)
   - If population is $\mathcal{N}(\mu, \sigma)$ with unknown $\sigma$, the sampling distribution of $\overline{x}$ can be determined using the fact that $(\overline{x} - \mu)/(s/\sqrt{n})$ is $t_{n-1}$. (Again, there is not common notation that allows us to say, for example, $\overline{x}$ has a sampling distribution of $t_{n-1}(\mu, \sigma)$.)

4. Choose an alternative hypothesis, $H_a$. For example $\mu \neq 1.0$ or $\mu < 0.5$.

5. Choose a rejection region. If the value of $\hat{\theta}$ from the sample, $\hat{\theta}_{\text{sample}}$, is in this region, the equality claim will be rejected. The rejection region has associated with it an area, $\alpha$. If $H_a$ is $\theta \neq \theta_0$, the rejection region consists of values of $\hat{\theta}$ in the lower $\alpha/2$ and upper $\alpha/2$ regions of the sampling distribution.

   If we assume population is $\mathcal{N}(\mu = 0, \sigma = 1)$ and execute an experiment of drawing $n$ values from it and compute $\overline{x}_{\text{sample}}$, we reject the null that $\mu = 0$ if $\overline{x}_{\text{sample}}$ is in the "rare" region of the sampling distribution for $\overline{x}$. We do not reject the null hypothesis otherwise.

   Based on the assumptions about the population distribution and the fact that the sampling distribution of $\overline{x}$ is $\mathcal{N}(\mu = 0, \sigma = 1/\sqrt{n})$ if these assumptions are true, we expect $|\overline{x}_{\text{sample}}| > 1.96\sigma/\sqrt{n}$ to occur in 5% of hypothetical experiments.

   When we reject, we are claiming "if we could repeat the experiment many times and compute many $\overline{x}_{\text{sample}}$ values, values in my selected rejection region are rare, so I'll reject the hypothesis if an experiment yields a $\overline{x}_{\text{sample}}$ in the rejection region; otherwise, I won't reject the null hypothesis". Notice that the statements don't involve claims that $H_0$ is true or false. They only involve statements about whether the observation is likely or not.

In summary, a hypothesis test has the logic: Given a claim and the calculation of a statistic based on experimental observations, if the value of the statistic is unlikely based on what we know (or assume we know) about the probability of every possible statistic that could occur from an experiment, reject the claim; otherwise, don't reject the claim (it is tempting to say "accept the claim", but that is equivalent to making a statement that the claim is true).

See also https://online.stat.psu.edu/stat500/Lesson06

**Example**

The drying time of paint has historically been 75 minutes with a standard deviation of 9 minutes.

A new additive was introduced that is claimed to decrease the drying time.

$n = 25$ tests were done on the new paint, and the average drying time was 70.0 minutes. Assume the samples were drawn from a normally distributed population with a standard deviation of 9 minutes.

Give the test result and the hypothesis test

$H_0$: $\mu = 75$ and $H_a$ : $\mu < 70.9$,

1. Should $H_0$ be rejected?
2. What is the probability of a Type I error for this hypothesis test?
3. Suppose we wanted to change the rejection region to be $< 70.0$ so that we don't reject. What is $\alpha$ in this case?

*Answers*

1. Yes, because 70.0 is in the rejection region of $H_a$.
2. The rejection region is the area under the normal distribution from $z = -\infty$ to $z = Z$, where $Z = (70.8 - 75)/(9/\sqrt{25}) \approx -2.33$. This area is 0.01 and so this is $\alpha$, the probability of a Type I error.
3. $Z = (70.0 - 75)/(9/\sqrt{25}) \approx -2.78$. The area under the normal distribution from $z = -\infty$ to $z = -2.78$ is 0.0027 from a table. So $\alpha = 0.0027$. Later we will see how this extra calculation is related to performing a hypothesis test using $P$-values.

## 8.3 Interpretation

**Important**: "Rejecting $H_0$" does not mean $H_0$ is false – we don't know if it is true or false and in general never will. "Rejecting $H_0$" means $H_0$ is unlikely, where the unlikeliness threshold for rejection is determined by $\alpha$. In a similar way, a jury asserts "guilty" with the qualifier "beyond a reasonable doubt". In most cases, we will never know if the defendant is actually guilty. As a result, we use "guilty beyond a reasonable doubt" to mean that a jury thinks it is highly unlikely that their claim of "guilt" is wrong.

It is easy to get the wording wrong on a hypothesis test. I will probably do it when speaking. On Devore p. 301 he states "The objective [of a hypothesis test] is to decide, based on sample information, which of the two hypotheses is correct." I find "correct" could be misinterpreted as meaning "true", so I suggest a better version is "The objective of a hypothesis test is to compute, based on a sample, if a hypothesis is unlikely if the null hypothesis is true. This information can be used to determine if a hypothesis is incorrect with some level of confidence."

## 8.4 Relationship to CIs

> The conclusion drawn from a two-tailed confidence interval is usually the same as the conclusion drawn from a two-tailed hypothesis test. In other words, if the 95% confidence interval contains the hypothesized parameter, then a hypothesis test at the 0.05 level will almost always fail to reject the null hypothesis. If the 95% confidence interval does not contain the hypothesized parameter, then a hypothesis test at the 0.05 level will almost always reject the null hypothesis. (https://online.stat.psu.edu/stat200/lesson/6/6.6)

A confidence interval can be framed as a hypothesis test (See also Bulmer p 169). Consider the test

- $H_0$: $\theta = \theta_o$
- $H_a$: $\theta \neq \theta_0$

1. Test statistic: $\hat{\theta}$, which is an unbiased estimate of $\theta$.
2. Rejection region: $\hat{\theta}$ of sample outside the CI *centered on* $\theta_o$.

If $\theta_{\text{sample}}$ is in rejection region, we reject $H_0$.

> Confidence intervals have recently been promoted over the use of hypothesis tests for a litany of unsupported reasons. (a) Among its supposed benefits is the assertion that confidence intervals provide more confidence than do hypothesis tests. This is based on the fallacy that confidence intervals are based on some system of probability theory other than that of hypothesis tests, when in fact they are the same. (b) Another prevalent misconception is confidence intervals must be symmetric. (Sawilowsky, 2011, Statistical Fallacies)

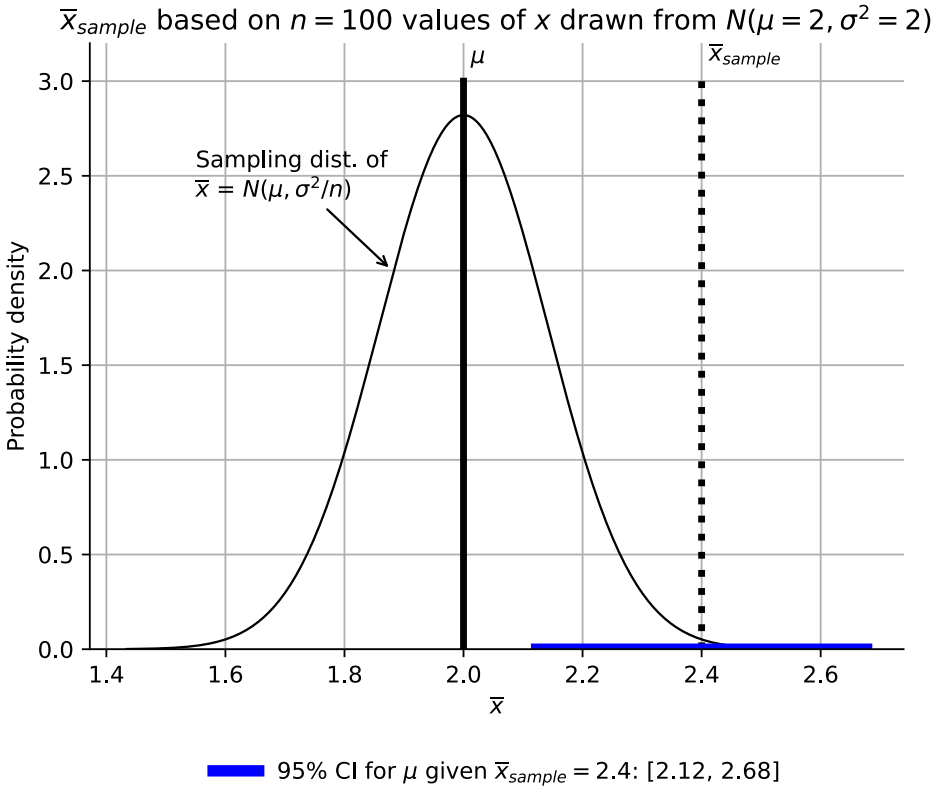
**Example**

Consider the following experiment: you are given a sample of $n = 100$ values. You are told that they were generated by a Gaussian random number generator with mean $\mu = 2$ and variance $\sigma^2 = 2$. You are certain about the Gaussian and $\sigma^2$ claim (this seems unrealistic, but simplifies the analysis), but suspect the $\mu = 2$ claim is wrong because you find $\overline{x}$ of the sample is 2.4.

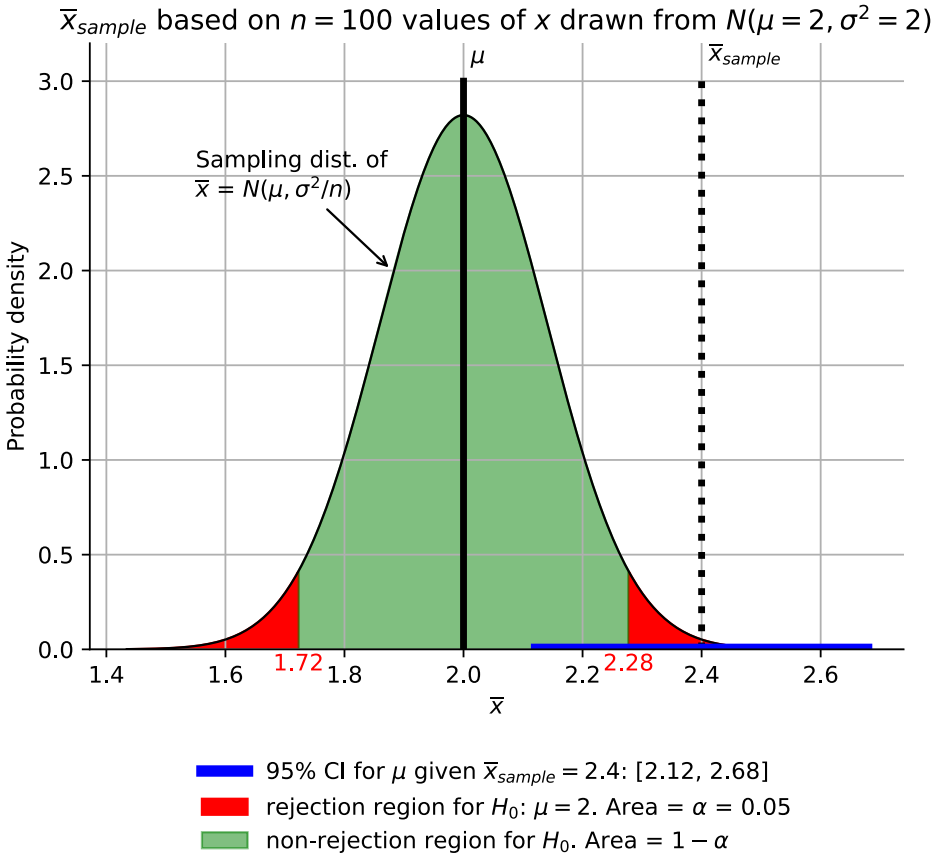
1. Compute a 95% confidence interval for $\mu$.
2. Use a hypothesis test using the rejection region algorithm to assess the claim about $\mu$. Use a significance level of 5% and let $H_0$ be $\mu = 2$ and $H_a$ be $\mu \neq 2$.
3. Make an assessment about the chances that you reject $H_0$ and $H_0$ is true (Type I error)
4. Explain how the confidence interval and the results of parts 1. and 2. are related.

*Answers*

1. The appropriate CI for $\mu$ is $[\overline{x} - z_{0.025}\sqrt{2}/\sqrt{100}, \overline{x} + z_{0.025}\sqrt{2}/\sqrt{100}]$, where $z_{0.025} = 1.96$. The values are given in the figure below along with a visualization of this 95% confidence interval. The black curve is the sampling distribution of $\overline{x}$ assuming that samples were drawn from $\mathcal{N}(\mu, \sigma^2)$. $\overline{x}_{\text{sample}}$ is the actual value found from a sample of $n$ values. If we were to repeat this process many times, we expect that in 95% of the cases, the blue line will overlap 2.0.




2. We reject $H_0$ if the observed value is in the red region. The red regions can be determined by shifting the blue line to be centered on $\mu$.




3. $\alpha = 0.05$.

## 8.5 Basic $P$–Value Algorithm

> The $P$-value is the probability, calculated assuming that the null hypothesis is true, of obtaining a value of the test statistic at least as contradictory to $H_0$ as the value calculated from the available sample. (Devore p. 329)

To find a $P$-value, modify the $\alpha$ associated with the rejection region so that it is at the threshold of where one would reject the null.

> $H_0$ should be rejected in favor of $H_a$ when the $P$-value is sufficiently small. (Devore p. 330)

The basic $P$ value algorithm is a generalization of the basic rejection region method. Instead of simply computing if an observed value is in a rejection region, we compute how far the observed value is in the rejection region (or how far away from the rejection region boundary it is).

Instead of rejecting based on a predefined rejection region, we reject based on a requirement for the probability of a Type I error (which we have denoted as $\alpha$) if we change $\alpha$ to be at the threshold of rejection

of $H_0$.

**Example**

This example was considered in the Basic Rejection Region Algorithm section.

The drying time of paint has historically been 75 minutes with a standard deviation of 9 minutes.

A new additive was introduced that is claimed to decrease the drying time.

$n = 25$ tests were done on the new paint, and the average drying time was 70.0 minutes. Assume the samples were drawn from a normally distributed population with a standard deviation of 9.

The hypothesis test of $H_0$: $\mu = 75$ with a rejection region of $< 70.8$ was considered and $H_0$ was rejected because the sample value of 70.0 is in the rejection region. It was found that this rejection region corresponded to $\alpha = 0.01$. The equivalent hypothesis using a $P$–value is

$H_0$: $\mu = 75$ with $H_a$: $\mu < 75$ and reject if $P \leq \alpha = 0.01$

Previously, we computed that if we wanted to change the rejection region to be $< 70.0$ (a threshold based on the observed value) so that we don't reject, we needed $\alpha = 0.0027$. This is the observed $P$ value. Since this $P$ value is less than the $P$-value specified in the hypothesis test (0.01), we reject.

Now we can state "reject $H_0$; $P = 0.0027$" or "$\mu < 75$; $P = 0.0027$."

Ideally we would state what the rejection threshold is for $P$, which was 0.01:

$H_0$: $\mu = 75$ with $H_a$: $\mu < 75$ and reject if $P \leq 0.01$

$P = 0.0027$ was found $\Rightarrow$ reject $H_0$.

The threshold is often omitted when $P$ is less than 0.01.

When using the rejection region algorithm, we only stated "reject $H_0$" with a significance level of 0.01. So the $P$ value method provides more information. It tells us how much we would need to adjust $\alpha$ to change our decision to reject or not reject $H_0$.

**Problem**

Rephrase the hypothesis test and the statement of its result for the example in Relationship to CIs section.

## 8.6 More Interpretation

> The rejection of a hypothesis at the 5% level does not imply that the probability that the hypothesis is false is 95%; it merely implies that the observed result belongs to a class of results whose overall probability of occurrence, if the null hypothesis is true, is 5%. This provides good reason, in the sense of a rational degree of belief, for supposing the hypothesis to be false, but no numerical value can be placed upon this degree of belief. (Bulmer, p. 165)

See also Statistical tests, P values, confidence intervals, and power: a guide to misinterpretations

## 8.7 $\beta$ and Power

(Only briefly covered. Will not be on Midterm.)

Recall

> A type I error consists of rejecting the null hypothesis $H_0$ when it is true.
>
> A type II error involves not rejecting $H_0$ when $H_0$ is false.
>
> The choice of a particular region cutoff values fixes the probabilities of type I and type II errors. These error probabilities are traditionally denoted by $\alpha$ and $\beta$, respectively.
>
> (Devore p. 304)

Thus far, the probability, $\alpha$, of a Type I error has been computed.

The probability of a type II error requires more computation because $H_0$ can be false in many ways.

For example $H_0$: $\mu = 10$ can be false if $\mu$ has any other value than 10, say $\mu_a$. So we need to find $\beta$ as a function of all possible values of $\mu_a$.

# 9 Regression

This is well covered in Devore and Bulmer.

## 9.1 General Remarks

See also Bulmer Chapter 12.

The 1–D regression model is

$y_i = \beta_0 + \beta_1 x_i + \epsilon_i$

This model is generally used for two purposes:

1. Given that there is a theoretical explanation for why there is a linear dependence of $y$ on $x$, estimate the parameters $\beta_0$ and $\beta_1$. For example, suppose we are told an object is moving with a constant but unknown velocity. We take measurements of its position $y$ at various times $t$. Then we have a list of $(y, t)$ pairs. A scatter plot should show the points along a straight line, the slope of which is $v$. We can use regression to find $v$. (We could use only two $(y, t)$ pairs to estimate $v$, but regression gives us a way to use more values to get a more accurate estimate.)
2. Give two variables $y$ and $x$, find how they co–vary. Here we may not have a physical equation that predicts the relationship between the variables, but seek to see how they are related (or associated). For example, we could measure eye color and height and ask if there is an association and characterize this by a linear fit. We don't have a physical equation that justifies the linear relationship and so the coefficients of the fit cannot be compared to theory. In this type of problem, we are more interested in if there is an association and this association is typically quantified by the correlation coefficients (or its square) and a confidence interval (or a hypothesis test that the correlation coefficient is not zero).

## 9.2 Derivation

For reference, in class I derived the least squares estimate for

$y = b_0 + b_1 x$

by minimizing the sum of squared errors

$$\text{SSE}(b_0, b_1) = \sum_{i=1}^{n} \left[ y_i - (b_0 + b_1 x_i) \right]^2$$

by using (limits on sum dropped to simplify)

$$\frac{\partial \text{SSE}}{\partial b_0} = 0 = -2 \sum (y_i - b_0 - b_1 x_i) = -2 \sum y_i + 2nb_0 + 2b_1 \sum x_i = -2n\overline{y} + 2nb_o + 2b_1 n\overline{x}$$

and

$$\frac{\partial \text{SSE}}{\partial b_1} = 0 = -2 \sum x_i y_i + 2b_0 \sum x_i + 2b_1 \sum x_i^2$$

and finding $b_0$ and $b_1$.

The first equation gives

$b_0 = \overline{y} - b_1 \overline{x}$

Substituting this into the second equation gives

$$b_1 = \frac{\left( \sum_{i=1}^{n} x_i y_i \right) - n\overline{xy}}{\left( \sum_{i=1}^{n} x_i^2 \right) - n\overline{x}^2} = \frac{\sum_{i=1}^{n} (x_i - \overline{x})(y_i - \overline{y})}{\sum_{i=1}^{n} (x - x_i)^2}$$

A faster way to get at the second form is to write

$$\text{SSE}(b_0, b_1) = \sum_{i=1}^{n} \left[ y_i - (b_0 + b_1 x_i) \right]^2 = \sum_{i=1}^{n} \left[ y_i - (\overline{y} - b_1 \overline{x} + b_1 x_i) \right]^2 = \sum_{i=1}^{n} \left[ (y_i - \overline{y}) - b_1 (x_i - \overline{x}) \right]^2$$

and then solve for $b_1$ in

$$\frac{\partial \text{SSE}}{\partial b_1} = -2 \sum \left[ (y_i - \overline{y}) - b_1 (x_i - \overline{x}) \right] (x_i - \overline{x}) = 0$$

## 10 Multicollinearity

The 1–D regression model is

$y_i = \beta_0 + \beta_1 x_i + \epsilon_i$

The 2–D regression model is

$y_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \epsilon_i$

Problems occur if $x_{1i}$ and $x_{2i}$ are correlated (called "multicollinearity"). Suppose we take photos of an object moving at a constant velocity and write down a table of

```
y      t      n (photo #)
1.0    0.0       1
2.1    0.1       2
3.0    0.2       3
4.1    0.3       4
```

Clearly $y$ depends on $t$ and $n$. So we may be tempted to try fitting

$y = b_0 + b_1 t + b_2 n$

with the justification that the more information the better. Doing the regression gives

$b_0 = -0.1000$

$b_1 = -0.0000$

$b_2 = 0.1000$

According to this, $y$ does not depend on time, $t$, because $b_1 \simeq 0$!

The problem is that $t$ and $n$ are related by $t = (n-1)/10$, so $n = 10t + 1$ and we can write

$y = b_0 + b_2 + b_1 t + b_2(10t + 1) = b_0 + b_2 + (b_1 + 10b_2)t$

Defining $b_0' = b_0 + b_2$ and $b_1' = b_1 + 10b_2$ gives

$y = b_0' + b_1' t$

If we perform the regression, we get $b_0' \simeq -0.0996$ and $b_1' \simeq 0.0979$.

---

On this web page, an example is given using real data.

The author, Jim, notes that the problems created by multicollinearity include:

- The coefficient estimates can swing wildly based on which other independent variables are in the model.
- The coefficients become very sensitive to small changes in the model.

Jim also notes that "Multicollinearity affects the coefficients and $p$-values, but it does not influence the predictions, precision of the predictions, and the goodness-of-fit statistics. If your primary goal is to make predictions, and you don't need to understand the role of each independent variable, you don't need to reduce severe multicollinearity."

Devise a numerical experiment using data that you create that demonstrates the above statements by Jim.

## 11 DFT

The Fourier series model for $y$ having uniformly spaced in time values $y_0, y_1, ..., y_{N-1}$ is

---

**For $N$ odd**:

$$y_t = \alpha_0 + \sum_{i=1}^{q} [\alpha_i \cos(2\pi f_i t) + \beta_i \sin(2\pi f_i t)] + \epsilon_t \,,$$

The least-squares estimates for the $\alpha$ and $\beta$ values (the estimates for $\alpha_i$ are $a_i$ and the estimates of $\beta_i$ are $b_i$):

$$a_0 = \frac{1}{N}\sum_{t=0}^{N-1} y_t \equiv \overline{y} \qquad a_i = \frac{2}{N}\sum_{t=0}^{N-1} y_t \cos(2\pi f_i t) \qquad b_i = \frac{2}{N}\sum_{t=0}^{N-1} y_t \sin(2\pi f_i t)$$

with $i = 1, 2, ..., q$, $q = (N-1)/2$, and $f_i = i/N$.

There are a total of $N$ unknown parameters in the model equation. If $N$ is odd, for example $N = 5$, then $q = 2$ and the unknowns are $a_0, a_1, a_2, b_1, b_2$.

---

---

**For $N$ even**:

$$y_t = \alpha_0 + \sum_{i=1}^{q} \alpha_i \cos(2\pi f_i t) + \sum_{i=1}^{q-1} \beta_i \sin(2\pi f_i t) + \epsilon_t \,,$$

The least-squares estimates for the $\alpha$ and $\beta$ values (the estimates for $\alpha_i$ are $a_i$ and the estimates of $\beta_i$ are $b_i$):

$$a_0 = \frac{1}{N}\sum_{t=0}^{N-1} y_t \equiv \overline{y} \qquad a_i = \frac{2}{N}\sum_{t=0}^{N-1} y_t \cos(2\pi f_i t) \qquad a_{q+1} = \frac{1}{N}\sum_{t=0}^{N-1} y_t \cos(2\pi f_i t)$$

$$b_i = \frac{2}{N}\sum_{t=0}^{N-1} y_t \sin(2\pi f_i t)$$

with $i = 1, 2, ..., q$, $q = N/2 - 1$, and $f_i = i/N$.

There are a total of $N$ unknown parameters in the model equation. If $N$ is even, for example $N = 4$, then $q = 1$ and the four unknowns based on the above are $a_0, a_1, a_2, b_1, b_2$.

---

The values of $a$ and $b$ are also called the discrete Fourier transform (DFT) coefficients.

(The equations for the parameter estimates $a_i$ and $b_i$ are found by multiplying the Fourier series model equation by $\cos(2\pi f_i t)$ and $\sin(2\pi f_i t)$ and then summing both sides over $t = 1$ to $N$. In the E&M book by Griffiths, he calls this “Fourier’s trick”.)